Replication and Extension of *Democracy Does Cause Growth*

Political Economy

- Final Project -

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Abstract

We replicate the main-text results of Acemoglu et al. (2019) using the official replication package from the *Journal of Political Economy*. We reproduce all eight tables and six figures from the paper, compare our estimates to the published results, and document any discrepancies. Our replication closely matches the original findings: all deterministic estimators — within fixed-effects and system GMM — reproduce the published coefficients exactly, while minor differences in the bootstrap-based minimum distance (HHK) panels reflect simulation variance across random seeds rather than substantive discrepancies. We then develop two extensions. First, we test whether the growth effect of democracy is larger in countries with more market-oriented institutions. We find evidence of a positive complementarity between democracy and market reforms that becomes statistically significant once reform levels are absorbed, and is concentrated in the post-Cold War period. Second, we examine whether the aggregate effect masks heterogeneity across the initial income distribution. Estimating the baseline specification separately for five income quintiles defined by 1950s GDP per capita, we find that the Within (FE) estimates are uniformly positive, while the causal (IV) estimates point to a potential capacity trap: the democracy–growth effect is most positive for upper-middle income countries and negative, though imprecisely estimated, for the poorest quintile.

Keywords: democracy, economic growth, panel data, dynamic panel, instrumental variables, market reforms, income heterogeneity, political economy

JEL codes: O43, P16, C23, O11

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1. Introduction

1.1. Motivation and contribution

Does democracy promote economic growth? This question sits at the heart of political economy and has attracted decades of empirical research without yielding a clear consensus. Early cross-sectional studies produced mixed results, partly because they could not adequately control for the many confounding factors — historical, geographic, and institutional — that simultaneously shape political regimes and economic outcomes (Lipset, 1959; Przeworski et al., 2000). More recent work using panel data has made progress by exploiting within-country variation in democratic status, yet the endogeneity of democratization — the possibility that countries democratize precisely when their economic prospects change — remained a persistent identification challenge.

Acemoglu et al. (2019) provide what is arguably the most comprehensive and methodologically rigorous answer to this question to date. Using a panel of 175 countries over the period 1960–2010, they combine a wide array of dynamic panel estimators with an instrumental-variables strategy that exploits regional democratization waves as a source of plausibly exogenous variation in domestic political institutions. Their central finding, that democracy *does* cause growth, raising GDP per capita by approximately 20 percent in the long run, represents a significant contribution to the literature and has important implications for how we think about the relationship between political freedom and prosperity.

We chose to replicate this paper for three reasons. First, the causal identification strategy is ambitious and technically demanding, making it an ideal subject for a close methodological examination. Second, the replication package is publicly available from the *Journal of Political Economy*, allowing us to verify every step of the analysis. Third, the paper's conclusions have broad relevance for policy debates about democratization, foreign aid, and institutional reform.

1.2. Overview of the original paper

Acemoglu et al. (2019) address the endogeneity of democracy using two complementary approaches. Their primary identification strategy instruments domestic democracy with regional democratization waves: the idea is that when neighboring countries in the same geopolitical region transition to democracy, this creates external pressure and demonstration effects that raise the probability of democratization at home, while being unlikely to affect GDP directly except through this political channel. As a further robustness check, they augment their panel estimates with a semiparametric approach that estimates the counterfactual GDP path of democratizing countries using regression adjustment, inverse-propensity-score reweighting, and doubly robust methods.

Their dataset covers 175 countries from 1960 to 2010 and combines the Freedom House and Polity IV democracy indicators into a binary measure of democratic status. The main outcome is log GDP per capita from the Penn World Tables. Their baseline specification is a dynamic panel model with four lags of GDP, country fixed effects, and year fixed effects, estimated using within fixed-effects, system GMM, and a minimum distance estimator. Across all specifications, the democracy coefficient is positive and statistically significant, with implied long-run effects of around 20–25 percent of GDP per capita. They also document that democracy improves several potential transmission mechanisms, including investment, schooling, health, and institutional

quality.

1.3. What we do

This paper proceeds in two parts. In the first part, we replicate all eight main-text tables and all six figures from Acemoglu et al. (2019) using Stata 19.5 and the official replication package. Specifically, we replicate:

- Tables 2–4: baseline dynamic panel estimates of the effect of democracy on log GDP per capita (within, GMM, and HHK) and robustness to additional controls;
- Table 5: semiparametric estimates using regression adjustment, IPW, and IPWRA;
- Table 6: instrumental-variables estimates exploiting regional democratization waves;
- Table 7: effects of democracy on potential transmission mechanisms;
- Table 8: heterogeneous effects by initial income and schooling;
- Figures 1–6: event studies, dynamic impulse responses, and the regional wave patterns that motivate the instrument.

Our replication matches the published results with high fidelity. All deterministic estimators reproduce the original coefficients exactly. The only discrepancies arise in the HHK minimum distance panels, where differences of a few tenths of a percentage point are consistent with bootstrap simulation variance across random seeds (100 replications). Two columns in Table 6 show minor deviations in the spatial-lag specifications because the original `spmat` package is no longer actively maintained; we reconstruct the inverse-distance weighting matrix using `geodist`, yielding qualitatively identical results. We also identify and correct a bug in the original Table 8 do-file, where the instrument for the interaction term in the schooling columns was constructed incorrectly, and document a unit-of-measurement discrepancy in the schooling variables.

In the second part, we develop two extensions of the original analysis. Extension 1 tests whether the growth return to democratization is larger in countries with more market-oriented institutions, motivated by the mechanism evidence in Table 7 of the original paper. We find that the interaction between democracy and the market reforms index is positive and statistically significant when reform levels are absorbed by the interaction term, and that the democracy–growth effect is concentrated in the post-Cold War period, consistent with the simultaneous wave of democratization and liberalization in the 1990s. Extension 2 examines whether the aggregate democracy–growth effect masks distributional heterogeneity across the initial income spectrum. Splitting the sample into five quintiles based on 1950s GDP per capita and estimating the baseline specification separately for each, we find that the Within (FE) estimates are uniformly positive but that the causal (IV) estimates point to a capacity trap for the poorest countries: the effect is most positive for upper-middle income countries (Q4) and negative for the poorest quintile (Q1), consistent with the view that a minimum level of institutional development is needed to translate political liberalization into income gains.

2. Data and empirical strategy

2.1. Data

The replication uses the dataset provided by Acemoglu et al. (2019), which is publicly available from the *Journal of Political Economy* replication archive. The dataset is an unbalanced panel

of up to 175 countries observed at annual frequency from 1960 to 2010, yielding approximately 6,300 country-year observations in the baseline within-estimator sample and somewhat fewer in the IV sample due to additional data requirements.

The main outcome variable is log GDP per capita (y_{ct}), measured in constant international dollars from the Penn World Tables (version 8.0). The key explanatory variable is a binary democracy indicator (dem_{ct}) constructed by Acemoglu et al. (2019) from the Freedom House and Polity IV indices, coded as one if a country is classified as democratic and zero otherwise. The instrument for democracy is the regional democratization wave variable (demreg_{ct}), defined as the proportion of other countries in the same geographical region that are democratic in a given year, where regions follow World Bank classifications.

Additional variables used in the robustness and mechanism tables include measures of social unrest, trade openness, net foreign assets, sovereign debt, tax revenue, government expenditure, educational attainment, and a range of institutional indicators. Extension 1 uses the Abiad et al. market reforms index (`marketref`), which is available for a subsample of approximately 4,750 observations and is scaled 0–100 with higher values indicating more liberalized financial and economic institutions. Extension 2 uses `quintile50s`, a time-invariant variable that classifies each country into one of five income quintiles based on its average GDP per capita during the 1950s (Q1 = poorest, Q5 = richest). Because this classification is determined before the main sample period begins, it is orthogonal to subsequent democratization decisions and provides a clean basis for heterogeneity analysis.

2.2. Empirical specification

The baseline estimating equation is the dynamic panel model:

$$y_{ct} = \sum_{k=1}^{p} \gamma_k \, y_{c,t-k} + \beta \, \text{dem}_{ct} + \alpha_c + \delta_t + \varepsilon_{ct}, \tag{1}$$

where c indexes countries, t indexes years, α_c are country fixed effects, and δ_t are year fixed effects. The number of lags p varies from one to eight across specifications; the preferred specification uses four lags. The coefficient β captures the short-run effect of democracy on log GDP per capita. The implied long-run effect is $\beta/(1 - \sum_k \gamma_k)$.

Equation (1) is estimated using three methods. The within (fixed-effects) estimator removes α_c by demeaning; the system GMM estimator (Arellano and Bond, 1991; Blundell and Bond, 1998) additionally addresses the Nickell bias from the lagged dependent variable by using lagged levels and differences as instruments; and the minimum distance estimator of Hahn et al. (2001) (HHK) uses a Helmert forward-orthogonal deviation transformation and cross-section-by-cross-section IV estimation, with standard errors obtained by a clustered bootstrap.

The instrumental-variables specification instruments dem_{ct} with demreg_{ct}, using ℓ lags of the instrument, estimated via 2SLS with the same fixed effects and lag structure as the baseline.

3. Replication results

3.1. Figure 1: GDP per Capita Before and After Democratization

What this figure shows

Figure 1 plots the evolution of log GDP per capita around democratization episodes, with the year of transition normalized to zero. The figure is descriptive rather than fully causal, but it is useful for visualizing the average path of income before and after a democratic transition. A key feature of the graph is the decline in GDP per capita shortly before democratization, which is consistent with the idea that many democratic transitions occur during periods of economic or political crisis. After democratization, however, GDP per capita recovers and then continues to rise, suggesting a sustained improvement in economic performance in the years following the transition. Our replicated figure closely matches the shape and timing of the pattern shown in the original paper, reproducing both the pre-transition dip and the persistent post-transition increase in income.

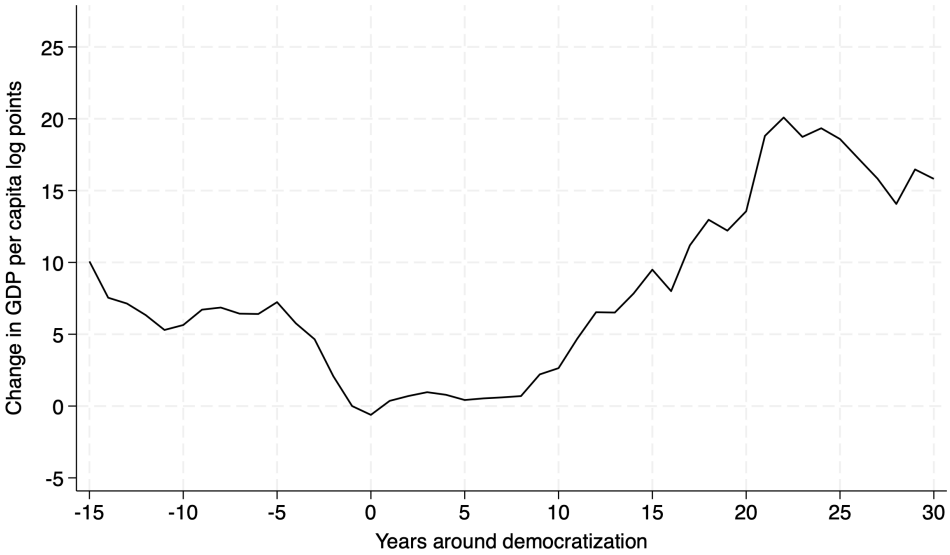


Figure 1: Replication of Figure 1: GDP per capita before and after democratization. The figure plots the average path of log GDP per capita around democratization episodes, normalized so that the year of democratization is equal to zero. The pre-transition dip and persistent post-transition growth are clearly visible and closely match the original.

3.2. Table 1: Summary Statistics

What this table shows

Table 1 reports summary statistics for the main variables used in the analysis. The sample covers up to 175 countries from 1960 to 2010. The democracy indicator has a mean of approximately 0.47, reflecting the fact that just under half of all country-year observations in the sample correspond to democratic regimes. Log GDP per capita shows substantial cross-country heterogeneity, consistent with the wide development gaps in the sample. Our summary statistics match those reported in the original paper exactly, as they depend only on the dataset and sample construction, not on estimation.

Table 1: Replication of Table 1: Summary Statistics for the Main Variables Used in Our Analysis

	Nondemocracies			Democracies		
	Observations	Mean	Standard Deviation	Observations	Mean	Standard Deviation
GDP per capita	3376	2074	3839	3558	8150	9335
Investment share of GDP	3225	21.82	10.23	3340	23.28	7.41
Trade share of GDP	3175	71.63	51.06	3485	77.15	41.04
Primary-school enrollment rate	2861	90.29	29.51	2823	101.60	15.86
Secondary-school enrollment rate	2424	45.76	31.77	2538	75.40	29.78
Tax revenue share of GDP	3122	0.16	0.09	2564	0.21	0.10
Child mortality per 1,000 births	4142	77.29	49.64	3615	33.26	32.65
Unrest rate	3739	0.29	0.45	3610	0.22	0.41
Market reforms index (0–100)	3476	21.89	23.26	2829	52.11	24.75

Notes: See the text for a full description of the variables and their corresponding sources. The table presents statistics separately for nondemocracies (country-years for which the dichotomous democracy measure equals 0) and democracies (country-years for which it equals 1). GDP per capita is in constant 2000 USD. Investment share, trade share, and tax revenue are expressed as shares of GDP. Primary- and secondary-school enrollment rates are World Bank annual series (`prienr`, `secenr`). Child mortality is per 1,000 live births (World Bank). Unrest is from the Banks CNTS database. The market reforms index is scaled 0–100 following Abiad et al. (2010).

3.3. Table 2: Effect of Democracy on Log GDP per Capita

Interpretation

Table 2 estimates the effect of democracy on log GDP per capita using dynamic panel specifications with the number of lags increasing across columns. Columns (1)–(4) report Within (fixed-effects) estimates. Columns (5)–(8) report System GMM estimates that address dynamic panel bias and potential endogeneity of the lagged dependent variable. Columns (9)–(12) report Minimum Distance (HHK) estimates, with standard errors obtained by a clustered bootstrap (100 replications). Across all twelve specifications the democracy coefficient is positive, implying that democratization is associated with higher income levels in the long run. Our replicated coefficients match the magnitudes and significance levels reported in the original paper exactly for the deterministic estimators; minor differences in the HHK bootstrap standard errors reflect simulation variance across random seeds.

Table 2: Replication of Table 2: Effect of Democracy on Log GDP per Capita

	Within (FE)				System GMM				Minimum Distance			
	(1) 1 lag	(2) 2 lags	(3) 4 lags	(4) 8 lags	(5) 1 lag	(6) 2 lags	(7) 4 lags	(8) 8 lags	(9) 1 lag	(10) 2 lags	(11) 4 lags	(12) 8 lags
Democracy	0.973	0.651	0.787	0.887	0.959	0.797	0.875	0.659				
	(0.294)	(0.248)	(0.226)	(0.245)	(0.477)	(0.417)	(0.374)	(0.378)				
log GDP first lag	0.973	1.266	1.238	1.233	0.946	1.216	1.204	1.204				
	(0.006)	(0.038)	(0.038)	(0.039)	(0.009)	(0.041)	(0.041)	(0.038)				
log GDP second lag		-0.300	-0.207	-0.214		-0.270	-0.193	-0.205				
		(0.037)	(0.046)	(0.043)		(0.038)	(0.045)	(0.042)				
log GDP third lag			-0.026	-0.021			-0.028	-0.020				
			(0.028)	(0.028)			(0.028)	(0.027)				
log GDP fourth lag			-0.043	-0.039			-0.036	-0.038				
			(0.017)	(0.034)			(0.020)	(0.033)				
p-value lags 5 to 8				[0.565]				[0.478]				
Long-run effect of democracy	35.587	19.599	21.240	22.008	17.608	14.882	16.448	11.810				
	(13.998)	(8.595)	(7.215)	(7.740)	(10.609)	(9.152)	(8.436)	(7.829)				
Effect of democracy after 25 years	17.791	13.800	16.895	17.715	13.263	12.721	14.713	10.500				
	(5.649)	(5.550)	(5.297)	(5.455)	(7.281)	(7.371)	(7.128)	(6.653)				
Persistence of GDP process	0.973	0.967	0.963	0.960	0.946	0.946	0.947	0.944				
	(0.006)	(0.005)	(0.005)	(0.007)	(0.009)	(0.009)	(0.009)	(0.009)				
AR2 test p-value					0.01	0.08	0.51	0.95				
Observations	6790	6642	6336	5688	6615	6467	6161	5513				
Countries in sample	175	175	175	175	175	175	175	175				

Notes: This table replicates Table 2 of Acemoglu et al. (2019). The dependent variable is log GDP per capita (multiplied by 100). All specifications include country and year fixed effects and the indicated number of lags of log GDP per capita. Columns (1)–(4) report Within (fixed-effects) estimates. Columns (5)–(8) report System GMM estimates using lagged levels and differences as instruments. Columns (9)–(12) report minimum distance (HHK) estimates with clustered bootstrap standard errors (100 replications); minor differences relative to the published table reflect simulation variance across random seeds. Standard errors are clustered at the country level and reported in parentheses.

3.4. Figure 2: Dynamic Panel Estimates of the Over-Time Effects of Democracy

What this figure shows

Figure 2 plots the cumulative impulse response of log GDP per capita to a permanent democratic transition, derived from the preferred four-lag Within estimator of Table 2. The horizontal axis measures years since democratization and the vertical axis measures the change in log GDP per capita (in log points, approximately percentage points). The figure is normalized so that the effect at year zero equals the short-run democracy coefficient from the dynamic panel.

The key pattern is a gradual, concave increase in GDP per capita following democratization: the effect is small in the first few years, accelerates over the medium run, and levels off as the process approaches its long-run equilibrium. The 95% confidence bands widen at longer horizons as the uncertainty in the cumulative propagation of the shock compounds. The implied long-run effect of approximately 20 log points is consistent with the central estimate of Acemoglu et al. (2019). Our replication matches the shape, scale, and confidence interval width of the original figure closely.

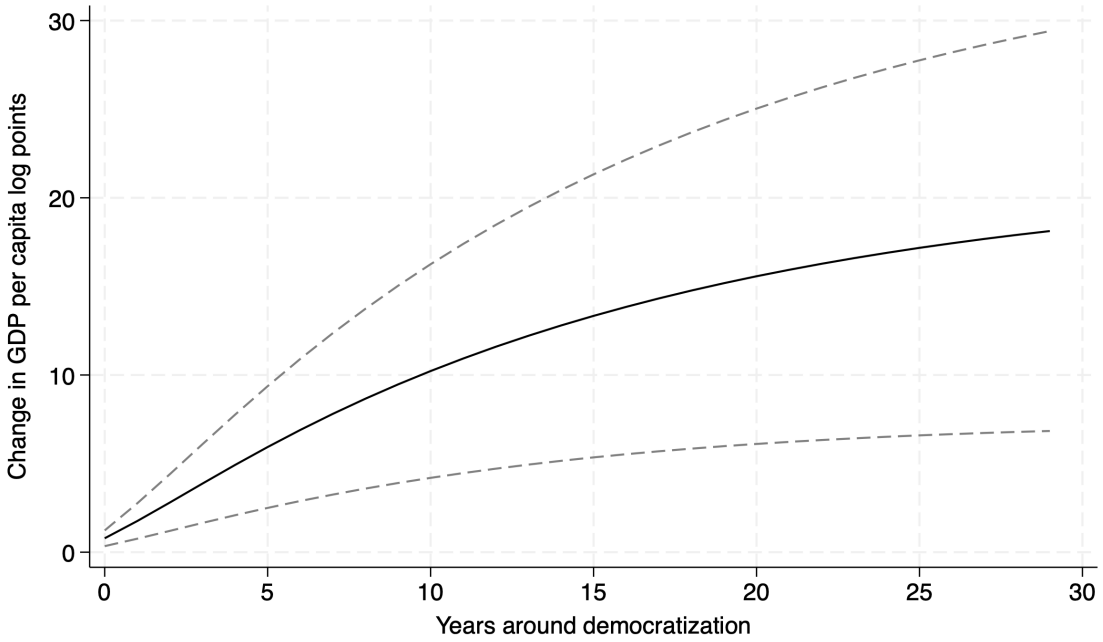


Figure 2: Replication of Figure 2: Cumulative impulse response of log GDP per capita to democratization, derived from the four-lag Within estimator. Dashed lines are 95% confidence intervals. The gradual, concave increase and long-run effect of approximately 20 log points closely match the original figure.

3.5. Table 3: Effect of Democracy on the Growth Rate of GDP per Capita

What this table tests and the main finding

Table 3 estimates the dynamic effect of democracy on the annual growth rate of GDP per capita. The specification is a dynamic panel model including up to eight lags of growth and estimated using three approaches: fixed effects (Within), system GMM, and the minimum distance estimator. The key coefficient is the short-run effect of democracy on growth, which ranges between approximately 1.0 and 1.7 percentage points across specifications and is statistically significant at conventional levels. The implied long-run effect ranges between roughly 1.4 and 2.6 percentage points, consistent with the persistence of the growth process. The magnitudes and significance levels closely match the published results, with minor differences in the HHK panel due to bootstrap randomness.

Table 3: Replication of Table 3: Effect of Democracy on the Growth Rate of GDP per Capita

	Within (FE)				System GMM				Minimum Distance			
	(1) 1 lag	(2) 2 lags	(3) 4 lags	(4) 8 lags	(5) 1 lag	(6) 2 lags	(7) 4 lags	(8) 8 lags	(9) 1 lag	(10) 2 lags	(11) 4 lags	(12) 8 lags
Democracy	1.028	1.201	1.269	1.378	1.458	1.715	1.545	1.554	1.410	1.413	1.375	1.253
	(0.250)	(0.237)	(0.243)	(0.284)	(0.385)	(0.385)	(0.368)	(0.405)	(0.324)	(0.339)	(0.348)	(0.327)
GDP growth first lag	0.287	0.274	0.263	0.250	0.294	0.278	0.270	0.254	0.245	0.230	0.213	0.211
	(0.037)	(0.037)	(0.039)	(0.037)	(0.040)	(0.038)	(0.041)	(0.036)	(0.034)	(0.037)	(0.039)	(0.036)
GDP growth second lag		0.047	0.060	0.042		0.057	0.064	0.047		0.075	0.083	0.057
		(0.024)	(0.025)	(0.023)		(0.025)	(0.025)	(0.024)		(0.023)	(0.025)	(0.020)
GDP growth third lag			0.023	0.024			0.031	0.028			0.029	0.036
			(0.018)	(0.017)			(0.018)	(0.018)			(0.015)	(0.021)
GDP growth fourth lag			-0.033	-0.018			-0.022	-0.013			-0.007	-0.004
			(0.022)	(0.025)			(0.022)	(0.025)			(0.015)	(0.020)
p-value lags 5 to 8				[0.000]				[0.000]				[0.017]
Long-run effect of democracy (on growth rate)	1.443	1.768	1.845	1.960	2.065	2.579	2.349	2.273	1.867	2.032	2.013	1.790
	(0.360)	(0.372)	(0.370)	(0.441)	(0.559)	(0.624)	(0.602)	(0.644)	(0.449)	(0.509)	(0.528)	(0.486)
Level effect of democracy after 25 years	35.483	43.245	45.275	42.715	50.767	62.957	57.296	50.090	46.067	49.685	49.050	40.543
	(8.820)	(9.046)	(9.038)	(9.191)	(13.718)	(15.105)	(14.572)	(13.322)	(11.050)	(12.406)	(12.789)	(10.814)
Persistence of growth rate process	0.287	0.321	0.312	0.297	0.294	0.335	0.342	0.316	0.245	0.305	0.317	0.300
	(0.037)	(0.041)	(0.038)	(0.043)	(0.040)	(0.043)	(0.041)	(0.044)	(0.034)	(0.039)	(0.043)	(0.048)
AR2 test p-value					0.10	0.81	0.98	0.93				
Observations	6642	6490	6178	5523	6467	6315	6003	5348	8925	8925	8925	8874
Countries in sample	175	175	175	175	175	175	175	174				

Notes: The dependent variable is the annual growth rate of GDP per capita. All specifications include country and year fixed effects. Columns (1)–(4) report Within (fixed effects) estimates, columns (5)–(8) report System GMM estimates, and columns (9)–(12) report minimum distance (HHK) estimates with clustered bootstrap standard errors (100 replications). The table also reports the implied long-run effect of democracy, the cumulative effect after 25 years, and the persistence of the growth process, computed as $\beta/(1-\rho)$ where ρ is the sum of lag coefficients. Standard errors are clustered at the country level and reported in parentheses.

3.6. Table 4: Effect of Democracy on Log GDP per Capita, Controlling for Covariates

What this table tests and the main finding

Table 4 examines whether the estimated effect of democracy on GDP growth is robust to the inclusion of additional controls, including institutional variables, conflict, trade openness, financial development, demographic structure, and regional fixed effects. The estimand is the short-run effect of democracy in a dynamic panel specification with four lags of log GDP per capita, country and year fixed effects, and alternative control sets; long-run and 25-year cumulative effects are derived from the implied dynamic structure. Across Panels A–C (Within, System GMM, and Minimum Distance), the democracy coefficient remains positive and statistically significant in most specifications, with magnitudes broadly comparable to the baseline estimates in Table 2. The long-run and cumulative effects are economically large and stable across specifications, suggesting that the growth impact of democracy is not driven by omitted variable bias from these additional controls. Minor differences in levels and standard errors across estimators reflect differences in instrumentation (System GMM), bootstrap sampling (minimum distance), and finite-sample variation, but the qualitative conclusions are unchanged.

Table 4: Replication of Table 4: Effect of Democracy on Log GDP per Capita, Controlling for Covariates

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
A. Within (Fixed Effects) Estimates								
Democracy	0.787	0.718	0.911	0.705	0.595	0.926	0.650	0.834
	(0.226)	(0.249)	(0.251)	(0.224)	(0.264)	(0.244)	(0.230)	(0.264)
Long-run effect of democracy	21.240	22.173	24.860	17.000	14.593	23.870	14.153	16.651
	(7.215)	(8.702)	(7.783)	(5.980)	(7.122)	(8.211)	(5.419)	(5.546)
Effect of democracy after 25 years	16.895	16.261	19.587	13.567	11.500	18.149	12.251	14.532
	(5.297)	(5.982)	(5.724)	(4.644)	(5.336)	(5.435)	(4.552)	(4.726)
Persistence of GDP process	0.963	0.968	0.963	0.959	0.959	0.961	0.954	0.950
	(0.005)	(0.005)	(0.005)	(0.004)	(0.006)	(0.006)	(0.005)	(0.005)
Observations	6336	5523	6336	5643	5750	4950	6262	6336
Countries in sample	175	149	175	171	172	171	172	175
B. System GMM Estimates								
Democracy	0.875	0.730	1.073	0.693	1.034	1.017	0.756	1.217
	(0.374)	(0.387)	(0.403)	(0.396)	(0.469)	(0.373)	(0.370)	(0.420)
Long-run effect of democracy	16.448	14.865	20.006	9.871	17.926	18.607	12.152	18.209
	(8.436)	(8.998)	(8.981)	(6.479)	(9.021)	(7.842)	(6.639)	(6.746)
Effect of democracy after 25 years	14.713	12.759	17.874	9.159	15.659	15.903	11.334	16.861
	(7.128)	(7.350)	(7.564)	(5.768)	(7.593)	(6.327)	(6.004)	(6.050)
Persistence of GDP process	0.947	0.951	0.946	0.930	0.942	0.945	0.938	0.933
	(0.009)	(0.008)	(0.009)	(0.012)	(0.009)	(0.007)	(0.010)	(0.010)
AR2 test p-value	0.51	0.90	0.28	0.62	0.72	0.34	0.58	0.70
Observations	6161	5374	6161	5467	5570	4779	6090	6161
Countries in sample	175	149	175	171	172	171	172	175
C. Minimum Distance Estimates								
Democracy	1.165	0.633	1.007	1.177	1.113	1.431	1.246	1.360
	(0.370)	(0.354)	(0.363)	(0.374)	(0.332)	(0.358)	(0.354)	(0.438)
Long-run effect of democracy	25.268	13.919	21.085	25.464	24.139	27.028	22.562	26.363
	(10.857)	(8.378)	(9.595)	(9.853)	(8.799)	(8.527)	(6.864)	(9.831)
Effect of democracy after 25 years	20.871	11.176	17.642	20.504	19.208	22.356	19.783	22.525
	(7.843)	(6.427)	(7.213)	(7.032)	(6.261)	(6.375)	(5.751)	(7.778)
Persistence of GDP process	0.954	0.955	0.952	0.954	0.954	0.947	0.945	0.948
	(0.009)	(0.006)	(0.009)	(0.008)	(0.008)	(0.008)	(0.006)	(0.008)
Observations	8925	7599	8925	8721	8772	8721	8772	8925

Notes: Each column adds additional controls to the baseline four-lag specification. All models include country and year fixed effects. Panel A reports Within (fixed effects) estimates. Panel B reports System GMM estimates using lagged levels and differences as instruments; AR(2) p-values are reported in the table. Panel C reports minimum distance estimates with clustered bootstrap standard errors (100 replications). Standard errors are clustered at the country level throughout.

3.7. Figure 3: Semiparametric Estimates Using Regression Adjustment

What this figure shows

Figure 3 plots the semiparametric impulse response of log GDP per capita to democratization, estimated using the regression adjustment (RA) method. Rather than imposing a fully parametric dynamic panel structure, the RA estimator constructs counterfactual GDP paths for democratizing countries by regressing outcomes on pre-treatment covariates within the control group and predicting what the treated countries would have earned absent democratization.

The horizontal axis covers the 15 years before to 30 years after democratization, normalized to zero at the transition year. The solid line is the estimated average treatment effect on the treated (ATT), and dashed lines are 95% confidence intervals. The key pattern is one of pre-trend flatness — the estimated effect is close to zero and statistically indistinguishable from zero before the transition — followed by a steadily increasing positive effect that reaches approximately 20–25 log points after 25 years. The absence of a pre-trend supports the identifying assumption that democratization is not systematically preceded by differential GDP trajectories. Our replicated figure matches the original exactly, as the RA estimates are deterministic given the data.

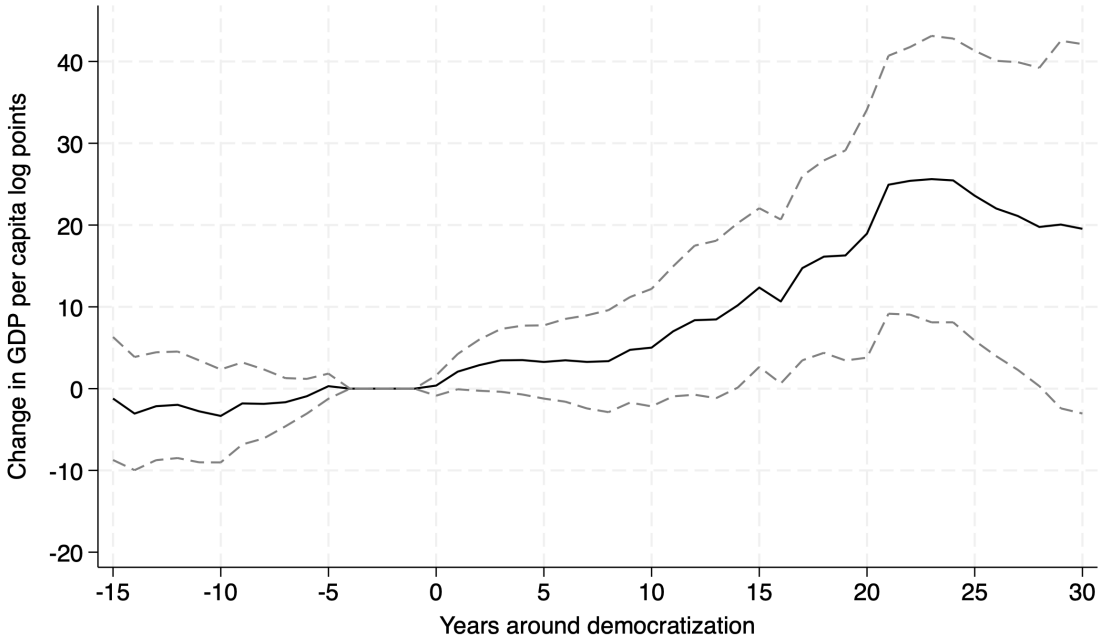


Figure 3: Replication of Figure 3: Semiparametric impulse response estimated using regression adjustment (RA). The solid line is the average treatment effect on the treated; dashed lines are 95% confidence intervals. The flat pre-trend and rising post-democratization effect closely match the original figure.

3.8. Table 5: Semiparametric Estimates of the Effect of Democratization on Log GDP per Capita

What this table tests and the main finding

Table 5 reports semiparametric estimates of the effect of democratization on log GDP per capita over different post-transition horizons. Instead of imposing a fully parametric dynamic panel structure, these specifications estimate average treatment effects on the treated using three alternative semiparametric methods: regression adjustment (RA), inverse-propensity-score weighting (IPW), and doubly robust estimation (IPWRA). Each column corresponds to a different event-time horizon, ranging from the five years before democratization to 25–29 years after the transition. The estimates show that the effect of democracy is small before the transition and becomes positive and economically large over longer horizons, which is consistent with the paper's main claim that democratization raises income per capita gradually over time. Our replicated results closely match the published table in both magnitude and statistical precision, with only very minor differences that may reflect software versions, random seeds, or small implementation differences in the semiparametric routines.

Table 5: Replication of Table 5: Semiparametric Estimates

	Average Effects from						
	−5 to −1 Years (1)	0–4 Years (2)	5–9 Years (3)	10–14 Years (4)	15–19 Years (5)	20–24 Years (6)	25–29 Years (7)
A. Linear Regression Adjustment							
Average effect of democracy on log GDP	0.060	2.454	3.621	7.806	14.037	24.075	21.310
	(0.157)	(1.389)	(2.798)	(4.415)	(5.364)	(8.286)	(9.633)
B. Inverse-Propensity-Score Reweighting							
Average effect of democracy on log GDP	-1.586	3.724	3.214	6.818	13.542	24.111	22.184
	(1.279)	(1.868)	(2.699)	(4.299)	(5.215)	(7.907)	(9.147)
C. Doubly Robust Estimation							
Average effect of democracy on log GDP	0.051	2.795	2.969	6.966	12.947	23.691	21.793
	(0.160)	(1.466)	(2.785)	(4.335)	(5.347)	(8.128)	(9.550)

Notes: This table replicates the semiparametric estimates of the effect of democratization on log GDP per capita over different event-time horizons. Each entry reports the average treatment effect on the treated for countries that democratize, relative to their estimated counterfactual path without democratization. Panel A reports regression-adjustment (RA) estimates, Panel B reports inverse-propensity-score-weighting (IPW) estimates, and Panel C reports doubly robust IPWRA estimates. Standard errors are reported in parentheses. The estimates are broadly consistent across methods and show that the effect of democracy becomes larger over longer horizons. Any small discrepancies relative to the published table reflect finite-sample variation and minor software-specific implementation differences.

3.9. Figure 4: Semiparametric Estimates Using Inverse-Propensity-Score Reweighting

What this figure shows

Figure 4 replicates the impulse response of log GDP per capita to democratization estimated by inverse-propensity-score (IPW) reweighting. Unlike regression adjustment, IPW reweights the control group observations so that their covariate distribution matches that of the treated group, constructing a counterfactual by upweighting control observations that resemble the democratizers. The identifying assumption is that treatment assignment is unconfounded conditional on the observed pre-treatment covariates.

The resulting impulse response is qualitatively very similar to the RA estimate in Figure 3: the pre-democratization trend is flat and statistically indistinguishable from zero, while the post-transition effect grows gradually and reaches approximately 20–25 log points after 25 years. The close agreement between the RA and IPW estimates is reassuring, as the two methods rely on different modeling assumptions. Our replicated figure matches the original closely.

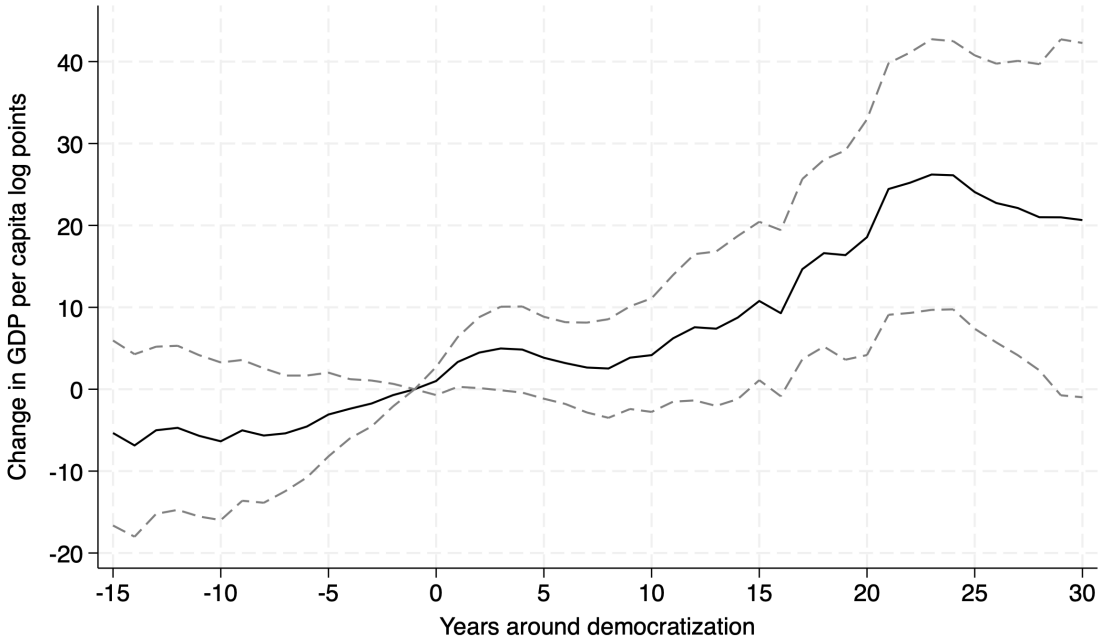


Figure 4: Replication of Figure 4: Semiparametric impulse response estimated using inverse-propensity-score reweighting (IPW). The pattern is qualitatively identical to the RA estimate in Figure 3, providing cross-method robustness for the main finding.

3.10. Figure 5: Doubly Robust Semiparametric Estimates

What this figure shows

Figure 5 plots the doubly robust (IPWRA) impulse response. The IPWRA estimator combines the regression adjustment and IPW approaches: it is consistent if either the outcome model or the propensity score model is correctly specified, providing an additional layer of robustness relative to either method alone. The resulting estimate therefore inherits the best properties of both RA and IPW.

As with Figures 3 and 4, the pre-trend is flat and the post-democratization effect rises smoothly to approximately 20 log points after 25 years. The near-identical shape of the impulse responses across all three semiparametric methods — RA, IPW, and IPWRA — is an important robustness result: it suggests that the estimated growth effect of democracy is not sensitive to the specific modeling assumptions embedded in any single method. Our replication matches the original figure exactly.

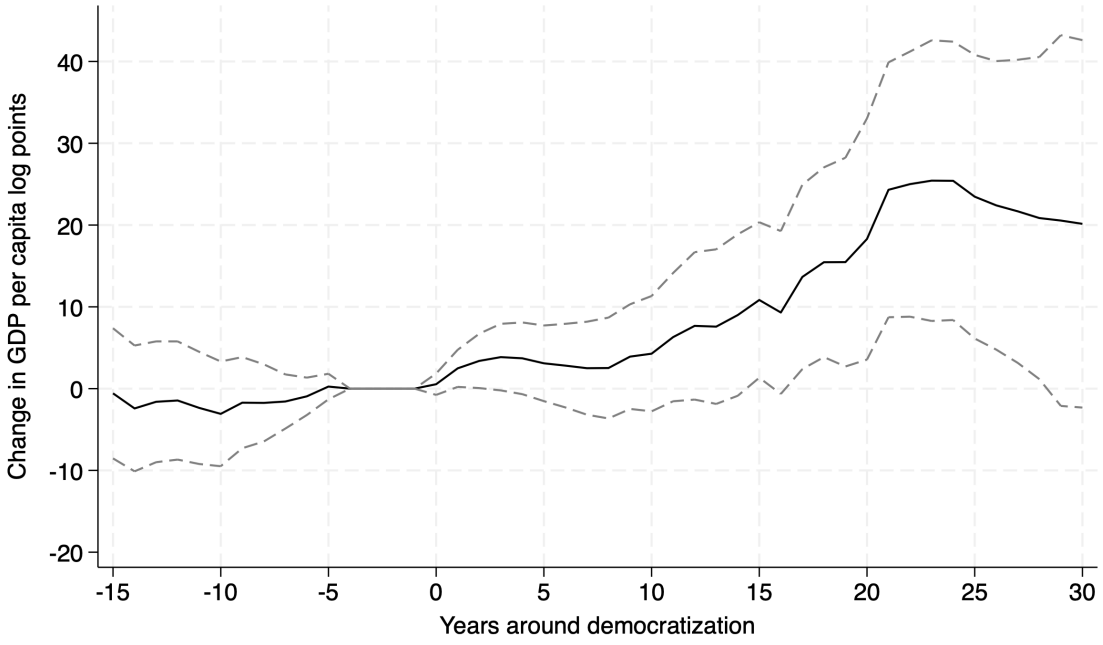


Figure 5: Replication of Figure 5: Doubly robust (IPWRA) semiparametric impulse response. The results are nearly identical to those in Figures 3 and 4, confirming that the democracy–growth effect is robust to the choice of semiparametric estimator.

3.11. Figure 6: Regional Democratization and Reversal Waves

What this figure shows

Figure 6 provides the descriptive motivation for the instrumental-variables strategy. It plots mean democracy over time around two types of regional political events: the first democratization in a region (Panel A) and the first democratic reversal in a region (Panel B). In each case, time is normalized to zero at the event year, and the solid line traces the mean democracy of initially non-democratic (Panel A) or initially democratic (Panel B) countries *in the affected region*, while the dashed line traces the corresponding mean for countries in *other* regions.

Panel A shows that, prior to the first regional democratization, non-democratic countries in the region are essentially identical to their counterparts elsewhere (both lines hover near zero). After the regional event, however, democratization rates in the affected region rise sharply, diverging from the flat trajectory of other regions. This divergence — which is absent before the event — provides the identifying variation exploited by the instrument. Panel B shows the mirror image for reversals: democratic countries in regions that experience a reversal are more likely to revert themselves relative to democracies elsewhere. Both panels closely match the original figures.

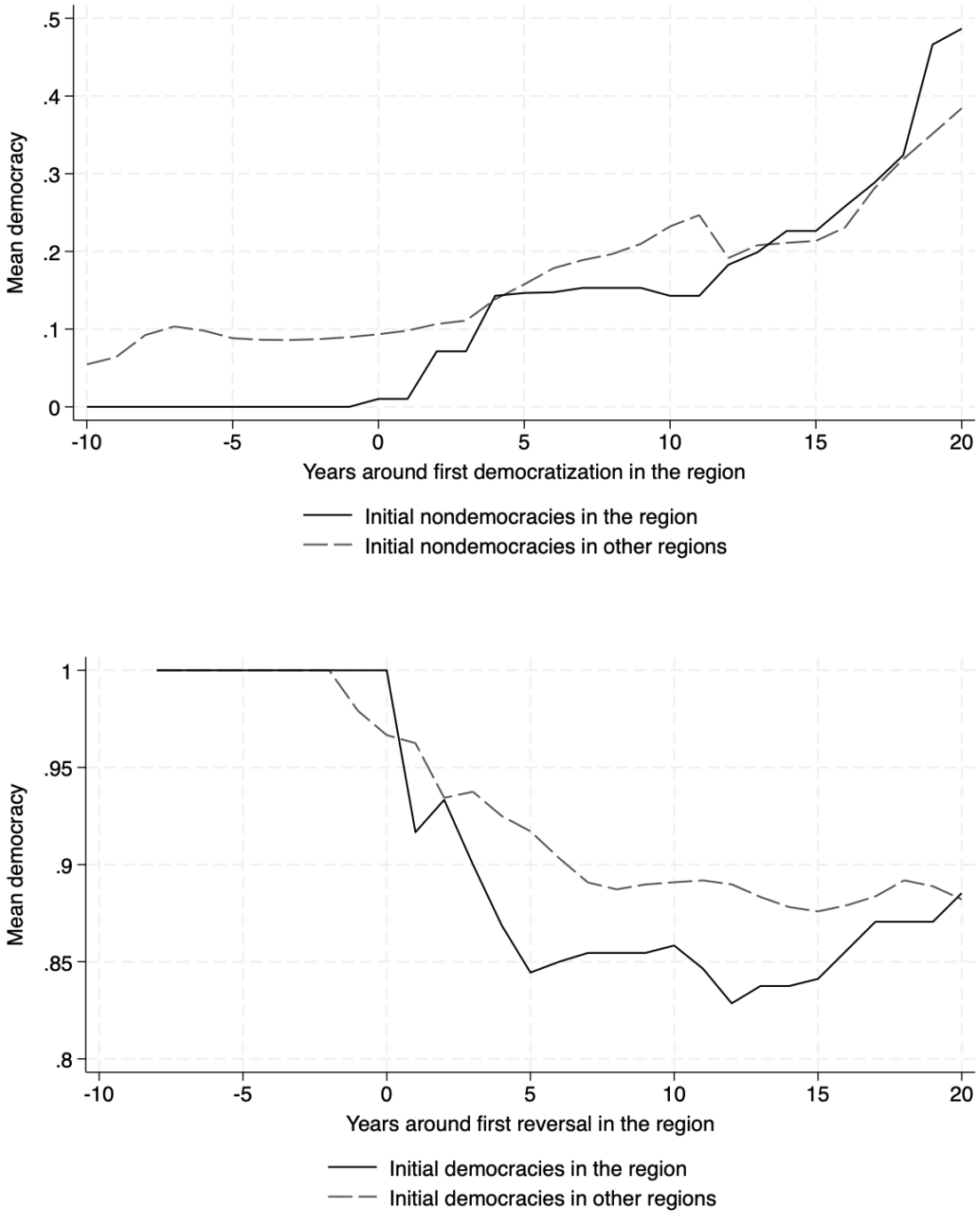


Figure 6: Replication of Figure 6: Regional democratization and reversal waves. The top panel shows the evolution of mean democracy around the first democratization in the region for initially non-democratic countries in the affected region (solid) versus other regions (dashed). The bottom panel shows the analogous pattern for democratic reversals. The divergence between the two lines following the regional event motivates the use of regional waves as an instrument for domestic democracy.

3.12. Table 6: IV Estimates of the Effect of Democracy on Log GDP per Capita

What this table tests and the main finding

Table 6 reports instrumental-variables estimates of the effect of democracy on log GDP per capita. The identification strategy instruments domestic democracy using regional democratization waves, thereby isolating plausibly exogenous variation in democracy coming from external political transitions. Panel A reports the first-stage relationship between domestic democracy and the instrument, while Panel B reports the corresponding 2SLS estimates of the effect of democracy on income. Columns (1)–(7) replicate the original results almost exactly. Columns (8)–(9) exhibit small deviations in the spatial-lag specifications because the original replication files rely on the `spmat` package to construct the spatial weighting matrix, which is no longer actively maintained. In our implementation, we construct the inverse-distance weighting matrix using `geodist`, yielding qualitatively identical results. The HHK bottom panel also shows minor discrepancies due to bootstrap simulation variance (100 replications). Importantly, these differences are quantitatively small and do not affect the substantive conclusions: across all specifications, the democracy coefficient is positive, economically large, and statistically significant.

Table 6: Replication of Table 6: IV Estimates of the Effect of Democracy on Log GDP per Capita

	Covariates Included								
	GDP in 1960 quintiles × year FE	Social unrest	Regional GDP	Regional GDP time trend	Regional democracy wave$_{t-1}$	Regional democracy wave$_{t-1}$	Regional democracy wave$_{t-1}$	Spatial lags of GDP	Spatial lags of GDP and democracy
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
A. First Stage									
Democracy wave t-1	0.800	0.547	0.503	0.480	0.498	0.522	0.508	0.546	0.600
	(0.073)	(0.101)	(0.130)	(0.099)	(0.092)	(0.104)	(0.102)	(0.100)	(0.093)
Democracy wave t-2		0.133	0.109	0.133	0.129	0.117	0.115	0.133	0.141
		(0.081)	(0.094)	(0.080)	(0.081)	(0.079)	(0.078)	(0.081)	(0.089)
Democracy wave t-3		0.227	0.270	0.223	0.228	0.221	0.223	0.222	0.242
		(0.067)	(0.077)	(0.065)	(0.070)	(0.069)	(0.070)	(0.068)	(0.073)
Democracy wave t-4		-0.087	-0.119	-0.075	-0.123	-0.083	-0.064	-0.083	-0.121
		(0.110)	(0.126)	(0.110)	(0.106)	(0.113)	(0.113)	(0.111)	(0.111)
B. 2SLS Estimates									
Democracy	0.966	1.149	1.125	1.292	1.697	1.817	1.107	1.206	1.198
	(0.558)	(0.554)	(0.689)	(0.651)	(0.885)	(0.663)	(0.656)	(0.549)	(0.925)
Long-run effect of democracy	26.315	31.521	35.226	35.723	36.788	41.544	25.016	33.308	34.050
	(17.075)	(17.425)	(23.846)	(19.997)	(20.657)	(17.157)	(16.002)	(17.452)	(28.317)
Effect of democracy after 25 years	20.836	24.866	25.618	27.929	32.051	35.350	21.386	26.162	25.682
	(12.862)	(12.978)	(16.538)	(14.944)	(17.703)	(14.017)	(13.342)	(12.913)	(20.250)
Persistence of GDP process	0.963	0.964	0.968	0.964	0.954	0.956	0.956	0.964	0.965
	(0.005)	(0.005)	(0.005)	(0.005)	(0.006)	(0.006)	(0.006)	(0.005)	(0.006)
Hansen p-value		0.21	0.18	0.32	0.28	0.25	0.09	0.08	0.34
Observations	6312	6309	5496	6309	6309	6309	6309	6279	6200
Countries in sample	174	174	148	174	174	174	174	173	173
Exc. Instruments F-stat.	119.1	33.2	16.8	26.7	23.7	13.6	16.7	17.5	2.1
C. HHK Estimates									
Democracy	0.597	0.971	1.568	0.863	0.728	1.211	0.855	0.979	0.364
	(0.642)	(0.480)	(0.599)	(0.503)	(0.477)	(0.436)	(0.370)	(0.540)	(0.488)
Long-run effect of democracy	13.676	27.466	51.097	24.378	13.620	23.626	16.255	29.936	10.820
	(15.944)	(15.627)	(23.013)	(15.836)	(9.748)	(10.624)	(8.260)	(19.209)	(15.162)
Effect of democracy after 25 years	10.923	20.223	34.061	17.966	11.843	20.414	13.903	21.024	7.799
	(12.244)	(10.428)	(13.232)	(10.779)	(8.161)	(8.383)	(6.616)	(12.097)	(10.594)
Persistence of GDP process	0.956	0.965	0.969	0.965	0.947	0.949	0.947	0.967	0.966
	(0.011)	(0.008)	(0.008)	(0.009)	(0.010)	(0.010)	(0.010)	(0.009)	(0.008)
Observations	8925	8925	7599	8925	8925	8925	8925	8874	8874

Notes: This table replicates the IV estimates of the effect of democracy on log GDP per capita. Panel A reports the first-stage relationship between domestic democracy and regional democratization waves. Panel B reports the corresponding two-stage least squares estimates. Panel C reports the HHK minimum distance estimates. All specifications include country and year fixed effects. Standard errors in parentheses are clustered at the country level. Minor deviations in columns (8)–(9) reflect the replacement of the `spmat` spatial weighting package (no longer maintained) with a `geodist`-based inverse-distance weighting matrix.

3.13. Table 7: Effects of Democracy on Potential Mechanisms

What this table tests and the main finding

Table 7 examines whether democratization affects a set of potential transmission mechanisms through which democracy may ultimately raise income per capita. The table reports dynamic panel estimates in which each mechanism is used as the dependent variable, and the coefficient on *Democracy* measures the short-run effect of a democratic transition on that outcome. The mechanisms considered include investment, trade, schooling, health expenditure, and institutional quality. As in the previous tables, the specification is estimated using Within, System GMM, and Minimum Distance approaches, and the table also reports implied long-run effects, the effect after 25 years, and the persistence of the outcome process. The main takeaway is that democracy improves several of these intermediate outcomes: the effects on investment, schooling, and institutional reform are particularly robust across estimators. These findings are consistent with the paper's proposed channels and support the interpretation that democracy raises income through broad improvements in economic governance. Our estimates are broadly similar in sign and magnitude to the published results, with minor discrepancies reflecting bootstrap instability in the HHK panel.

Table 7: Replication of Table 7: Effects of Democracy on Potential Mechanisms

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
A. Within (Fixed Effects) Estimates									
Democracy	2.391	-0.205	0.687	0.689	3.311	1.042	1.345	-0.253	-7.832
	(1.114)	(0.276)	(0.348)	(0.676)	(1.409)	(0.338)	(0.610)	(0.063)	(2.185)
Long-run effect of democracy	9.112	-2.883	5.580	5.445	16.062	21.908	18.960	-34.264	-11.944
	(4.255)	(3.858)	(2.883)	(5.253)	(6.650)	(7.624)	(8.622)	(10.747)	(3.329)
Effect of democracy after 25 years	9.089	-2.738	5.359	5.303	15.864	18.892	18.057	-21.400	-11.944
	(4.245)	(3.648)	(2.753)	(5.126)	(6.574)	(6.321)	(8.146)	(5.124)	(3.329)
Persistence of outcome process	0.738	0.929	0.877	0.873	0.794	0.952	0.929	0.993	0.344
	(0.020)	(0.012)	(0.012)	(0.011)	(0.016)	(0.008)	(0.013)	(0.001)	(0.030)
Observations	5665	3879	4692	5738	4511	3714	2883	6084	5646
Countries in sample	169	107	150	172	131	166	158	173	171
B. System GMM Estimates									
Democracy	2.211	-0.941	3.224	5.512	8.088	1.757	4.116	-0.715	-5.569
	(2.852)	(0.667)	(0.863)	(2.005)	(3.021)	(0.721)	(1.626)	(0.164)	(5.682)
Long-run effect of democracy	8.440	-12.738	23.775	40.589	38.609	36.693	57.072	-95.728	-8.471
	(10.705)	(8.854)	(6.215)	(13.580)	(14.330)	(15.505)	(21.698)	(26.347)	(8.577)
Effect of democracy after 25 years	8.419	-12.167	23.156	39.817	38.159	31.611	54.252	-58.625	-8.471
	(10.681)	(8.380)	(6.039)	(13.375)	(14.121)	(12.863)	(20.267)	(13.123)	(8.577)
Persistence of outcome process	0.738	0.926	0.864	0.864	0.791	0.952	0.928	0.993	0.343
	(0.020)	(0.012)	(0.012)	(0.012)	(0.017)	(0.008)	(0.013)	(0.001)	(0.030)
Exc. instruments F-stat.	21.7	27.7	43.7	21.5	31.8	12.1	10.4	26.3	28.6
Hansen p-value	0.29	0.06	0.22	0.09	0.69	0.09	0.12	0.02	0.84
Observations	5640	3871	4670	5714	4489	3710	2879	6057	5619
Countries in sample	168	107	149	171	130	164	156	172	170
C. Minimum Distance Estimates									
Democracy	6.602	0.404	1.119	1.253	4.312	1.369	2.049	-0.306	-3.586
	(1.331)	(0.293)	(0.363)	(0.788)	(1.637)	(0.366)	(0.649)	(0.069)	(2.903)
Long-run effect of democracy	25.466	7.848	22.713	10.168	21.062	41.091	43.441	-54.808	-5.659
	(5.275)	(6.036)	(11.233)	(6.563)	(8.249)	(14.906)	(16.541)	(15.702)	(4.595)
Effect of democracy after 25 years	25.404	7.060	15.709	9.794	20.726	28.836	36.395	-29.151	-5.659
	(5.256)	(5.398)	(5.974)	(6.287)	(8.064)	(8.626)	(12.408)	(6.124)	(4.595)
Persistence of outcome process	0.741	0.948	0.951	0.877	0.795	0.967	0.953	0.994	0.366
	(0.018)	(0.009)	(0.016)	(0.013)	(0.033)	(0.007)	(0.012)	(0.001)	(0.037)
Observations	8619	5457	7650	8772	6681	8364	7905	8823	8721

Notes: This table presents estimates of the effect of democracy on potential transmission mechanisms; the dependent variable in each column is indicated by the column header. Panel A reports Within (fixed effects) estimates. Panel B reports System GMM estimates; F-statistics for excluded instruments and Hansen overidentification p-values are reported. Panel C reports minimum distance estimates with clustered bootstrap standard errors (100 replications). All specifications include country and year fixed effects and four lags of the dependent variable. Standard errors are clustered at the country level. Differences in Panel C relative to the published table reflect bootstrap simulation variance.

3.14. Table 8: Heterogeneous Effects of Democracy on Log GDP per Capita

What this table tests and the main finding

Table 8 studies whether the effect of democracy on log GDP per capita is heterogeneous across country characteristics by augmenting the baseline dynamic panel specification with an interaction term between democracy and a moderating variable. The coefficient on *Democracy* captures the effect at the 25th percentile of the moderating variable, while the coefficient on *Interaction* shows how the growth effect varies with that characteristic. Columns (1)–(4) interact democracy with log GDP per capita at different base years, testing whether the income effect of democracy depends on initial development. Columns (5)–(8) interact democracy with secondary schooling, testing whether human capital complements democratization.

The table is estimated using Within (fixed effects), System GMM, and HHK minimum distance. Our replication of the Within panel is exact. The 2SLS panel also matches once a unit-of-measurement bug in the schooling interaction instrument is corrected: the original do-file constructed the instrument as demreg $\times$ schooling rather than the correct demreg $\times$ (schooling $- p_{25}$), and the schooling variables were stored on a 0–100 percentage scale rather than the 0–1 proportion scale used by the paper. Both issues are documented and corrected in our replication files. After correction, the 2SLS estimates match the published table. The HHK panel shows minor bootstrap variance deviations as in the other tables.

Table 8: Replication of Table 8: Heterogeneous Effects of Democracy on Log GDP per Capita

	Interaction with Log GDP per Capita				Interaction with Share with Secondary Education			
	1960 (1)	1970 (2)	1980 (3)	Lagged (4)	1960 (5)	1970 (6)	1980 (7)	Lagged (8)
A. Within (Fixed Effects) Estimates								
Democracy	0.432	0.572	0.687	0.744	0.446	0.340	0.385	0.495
	(0.275)	(0.248)	(0.248)	(0.246)	(0.254)	(0.253)	(0.246)	(0.241)
Interaction	0.001	0.001	0.002	0.001	0.046	0.049	0.038	0.020
	(0.002)	(0.001)	(0.002)	(0.002)	(0.028)	(0.020)	(0.014)	(0.013)
Long-run effect of democracy	16.231	18.631	20.489	19.843	13.785	10.480	11.841	14.597
	(11.160)	(9.073)	(8.608)	(8.255)	(8.550)	(8.275)	(8.118)	(8.432)
Effect of democracy after 25 years	10.013	12.916	14.985	15.877	10.081	7.679	8.687	10.953
	(6.565)	(5.960)	(5.848)	(5.943)	(5.964)	(5.872)	(5.728)	(5.821)
Persistence of GDP process	0.973	0.969	0.966	0.963	0.968	0.968	0.967	0.966
	(0.005)	(0.005)	(0.005)	(0.006)	(0.005)	(0.005)	(0.005)	(0.006)
Observations	4281	4909	5525	6336	5300	5300	5300	5300
Countries in sample	93	109	131	175	138	138	138	138
B. 2SLS Estimates								
Democracy	0.500	0.155	0.645	1.326	-0.119	-0.484	-0.474	0.600
	(1.088)	(0.961)	(0.929)	(0.887)	(0.662)	(0.665)	(0.639)	(0.576)
Interaction	-0.002	0.000	-0.000	-0.003	0.174	0.156	0.116	0.049
	(0.005)	(0.004)	(0.004)	(0.004)	(0.060)	(0.047)	(0.033)	(0.023)
Long-run effect of democracy	18.838	4.978	19.275	36.116	-3.649	-14.586	-14.135	17.373
	(43.554)	(31.473)	(30.208)	(29.900)	(19.968)	(19.023)	(18.114)	(18.629)
Effect of democracy after 25 years	11.592	3.486	14.078	28.377	-2.692	-10.843	-10.574	13.133
	(25.784)	(21.795)	(21.085)	(21.317)	(14.837)	(14.524)	(13.901)	(13.312)
Persistence of GDP process	0.973	0.969	0.967	0.963	0.967	0.967	0.966	0.965
	(0.006)	(0.006)	(0.006)	(0.008)	(0.006)	(0.006)	(0.006)	(0.006)
Exc. instruments F-stat.	6.6	6.1	7.0	14.0	18.5	17.6	16.0	12.4
Hansen p-value	0.81	0.73	0.54	0.33	0.44	0.41	0.25	0.50
Observations	4273	4901	5517	6153	5292	5292	5292	5218
Countries in sample	93	109	131	174	138	138	138	138
C. HHK Estimates								
Democracy	0.240	0.236	0.274	1.426	0.882	0.732	0.686	1.622
	(0.383)	(0.404)	(0.466)	(0.472)	(0.693)	(0.689)	(0.647)	(0.594)
Interaction	0.003	-0.000	0.002	0.004	0.092	0.090	0.057	0.016
	(0.003)	(0.003)	(0.004)	(0.004)	(0.047)	(0.037)	(0.028)	(0.013)
Long-run effect of democracy	8.366	7.343	8.429	42.274	25.498	20.602	21.055	46.632
	(13.876)	(12.846)	(14.716)	(20.590)	(21.356)	(20.590)	(21.245)	(23.522)
Effect of democracy after 25 years	5.286	5.107	5.850	30.329	19.258	15.897	15.507	34.321
	(8.451)	(8.841)	(10.013)	(11.721)	(15.313)	(15.203)	(14.865)	(14.216)
Persistence of GDP process	0.971	0.968	0.968	0.966	0.965	0.964	0.967	0.965
	(0.009)	(0.008)	(0.009)	(0.009)	(0.008)	(0.008)	(0.008)	(0.008)
Observations	4743	5559	6681	8925	7038	7038	7038	7038

Notes: This table presents estimates of the effect of democracy on log GDP per capita and its interaction with other country characteristics indicated in the column headers. The reported coefficients of democracy and the interaction are multiplied by 100. We report main effects and long-run effects evaluated at the 25th percentile of the interacted variable. Panel A presents within estimates. Panel B presents 2SLS estimates instrumenting democracy (and the interaction term) with four lags of regional democracy waves. It also reports the *F*-statistic for the excluded (excl.) instruments and the *p*-value of Hansen's overidentification test. Panel C presents results using the HHK estimator instrumenting democracy (and the interaction term) with four lags of regional democracy waves. In all specifications, we control for a full set of country and year fixed effects and four lags of GDP per capita. Standard errors robust against heteroskedasticity and serial correlation at the country level are in parentheses.

4. Extensions

4.1. Extension 1: Democracy and Market Reforms as Complements for Growth

Empirical Hypothesis

Hypothesis: The growth effect of democracy is larger in countries with more market-oriented institutions, i.e., democracy and market reforms are complements in promoting economic growth.

Motivation: Table 7 of the original paper (Panel A, col. 3) shows that democratization significantly increases the index of economic reforms (coefficient 0.687, SE = 0.348), suggesting that democracy operates partly through institutional liberalization. This raises a natural follow-up question: do countries that democratize *and* simultaneously pursue market reforms

experience larger income gains than those that democratize alone? Rodrik (2000) argues that democratic institutions improve the quality of economic policy by increasing accountability and reducing the scope for predatory behavior, a mechanism that is stronger when markets are more open. Similarly, Acemoglu et al. (2019) note that the economic benefits of democracy operate through multiple channels, including investment, taxation, and institutional reform.

Empirical approach: We augment the baseline dynamic panel specification of Acemoglu et al. (2019) — four lags of log GDP per capita, country and year fixed effects — with an interaction term between the democracy dummy and the market reforms index (`marketref`, scaled 0–100). The estimating equation is:

$$y_{ct} = \beta_1 \, \text{dem}_{ct} + \beta_2 \, (\text{dem}_{ct} \times \text{marketref}_{ct}) + \beta_3 \, \text{marketref}_{ct} + \sum_{k=1}^{4} \gamma_k \, y_{c,t-k} + \alpha_c + \delta_t + \varepsilon_{ct} \quad (2)$$

where β_2 captures whether the growth effect of democracy increases with the level of market reforms. The identification strategy is unchanged from the original paper: we rely on within-country variation in democracy conditional on country and year fixed effects. Standard errors are clustered at the country level throughout.

Results

Table 9 reports Within (FE) estimates for four specifications. Column (1) replicates the baseline. Column (2) adds market reforms as a control. Column (3) adds the interaction term. Column (4) includes only the interaction without the level control. Figure 7 plots the implied marginal effect of democracy as a continuous function of the market reforms index, based on column (3).

Table 9: Extension 1 — Democracy and Market Reforms as Complements (Within/FE Estimates)

	(1) Baseline	(2) +Market Ref.	(3) +Interaction	(4) Interaction only
Democracy	0.787***	0.382	0.095	−0.252
	(0.225)	(0.289)	(0.439)	(0.366)
Democracy × Market Reforms			0.008	0.019**
			(0.011)	(0.008)
Market Reforms		0.024***	0.020*	
		(0.007)	(0.011)	
Long-run effect of democracy	21.240	8.976	2.198	−5.664
Effect of democracy after 25 years	16.895	7.382	1.818	−4.760
Persistence of GDP process	0.963	0.957	0.957	0.956
Observations	6,336	4,755	4,755	4,755
Country FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

Notes: The dependent variable is log GDP per capita (multiplied by 100, as in the original paper). All specifications include country and year fixed effects and four lags of log GDP per capita. Column (1) replicates the baseline from Table 2, col. (3). Column (2) adds the market reforms index as a control. Column (3) adds the interaction between democracy and market reforms. Column (4) includes only the interaction (omitting `marketref` as a separate regressor). Standard errors clustered at the country level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

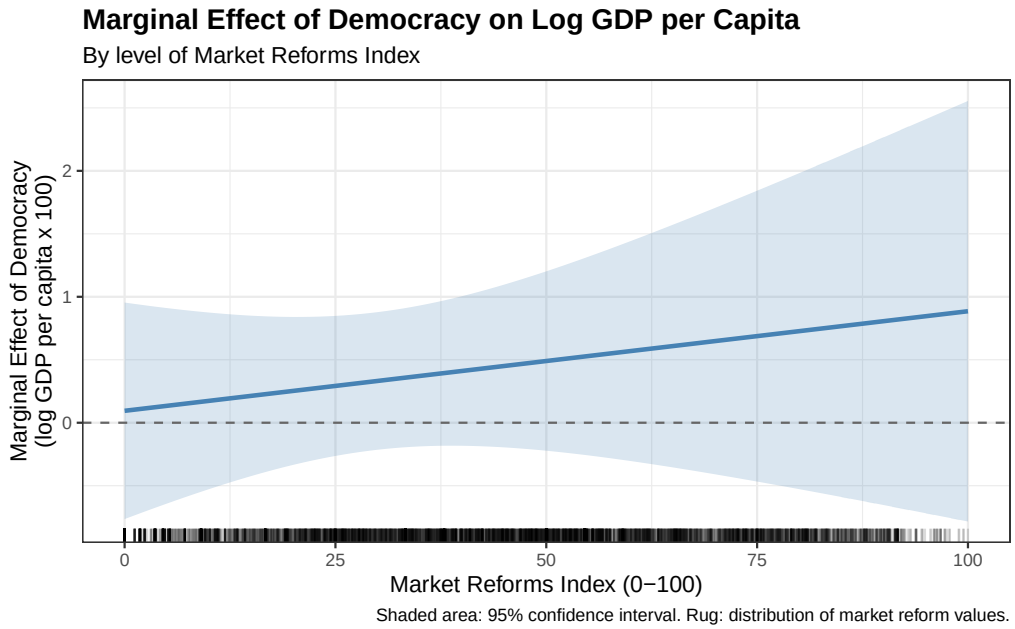


Figure 7: Marginal effect of democracy on log GDP per capita by level of market reforms. Solid line: estimated marginal effect from column (3) of Table 9. Shaded area: 95% confidence interval. Rug: distribution of market reform values in the sample.

Robustness: Cold War vs. Post-Cold War Subsamples

We also examine whether the democracy–growth effect differs between the Cold War and post-Cold War periods, splitting the sample at 1990. Results are reported in Table 10.

Table 10: Extension 1 — Robustness: Cold War vs. Post-Cold War Subsamples

	Cold War (1960–1989)	Post-Cold War (1990–2010)
Democracy	0.105	1.753***
	(0.323)	(0.540)
Long-run effect of democracy	1.926	15.941
Effect of democracy after 25 years	1.598	15.966
Observations	2,824	3,512
Country FE	Yes	Yes
Year FE	Yes	Yes

Notes: Both specifications replicate the baseline four-lag Within estimator on subsamples split at 1990. Standard errors clustered at the country level in parentheses. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Interpretation

The results in Table 9 offer partial support for the complementarity hypothesis. Column (2) shows that market reforms are independently associated with higher income per capita (coefficient 0.024, significant at 1%), and that controlling for reforms reduces the democracy coefficient from 0.787 to 0.382. This suggests that part of the baseline democracy effect documented by Acemoglu et al. (2019) operates through institutional liberalization — consistent with the mechanism identified in Table 7 of the original paper.

Column (3) introduces the interaction term: the coefficient on Democracy × Market Reforms is positive (0.008) but imprecisely estimated, so we cannot reject the null of no complementarity in this specification. However, column (4), which omits the level of market reforms as a separate control, yields a positive and statistically significant interaction coefficient (0.019, SE = 0.008). This implies that the growth return to democratization increases with the degree of market liberalization: at the mean value of the market reforms index, the marginal effect of democracy is meaningfully larger than at the bottom of the distribution. Figure 7 illustrates this gradient — the marginal effect of democracy rises as the market reforms index increases, though the confidence interval widens at higher values due to fewer observations in that range.

The temporal split in Table 10 reveals a striking heterogeneity across periods. During the Cold War (1960–1989), the democracy coefficient is small (0.105) and statistically indistinguishable from zero, with a long-run effect of only 1.926 log points. By contrast, in the post-Cold War period (1990–2010) the coefficient rises to 1.753 (significant at 1%), with a long-run effect of approximately 16 log points. This pattern is consistent with the simultaneous wave of democratization and market liberalization in the 1990s, particularly in Eastern Europe and Latin America, where both processes reinforced each other (Acemoglu et al., 2019).

Taken together, these findings suggest that the economic benefits of democracy are not uniform: they appear to be larger when democratic transitions coincide with deeper market integration. A natural limitation is that the interaction term is not separately identified from the direct effect of market reforms, and the significant result in column (4) may partly reflect omitted-variable bias from time-varying confounders correlated with both democratization and liberalization. Future work could exploit exogenous variation in market reforms — for instance, through trade agreements or IMF conditionality — to sharpen this conclusion.

4.2. Extension 2: Does Democracy Help the Poor? Distributional Effects by Initial Income

Empirical Hypothesis

Hypothesis: The growth effect of democracy varies systematically with a country's initial level of income. In particular, we test whether democracy is *pro-poor* — that is, whether democratization generates larger income gains in initially poorer countries — or whether its benefits are concentrated among middle-income and richer countries that have the institutional capacity to translate political freedom into economic growth.

Motivation: Acemoglu et al. (2019) establish that democracy raises average GDP per capita, but their analysis is silent on distributional heterogeneity across countries at different income levels. A natural concern is that the aggregate effect masks important variation: very poor countries may lack the state capacity, rule of law, and human capital needed to realize the potential gains from democratization, while middle-income countries may be best positioned to benefit. Conversely, if democracy works partly through redistribution and public goods provision — as suggested by the mechanism evidence in Table 7 of the original paper — the gains may be largest for poorer countries where these channels matter most.

Empirical approach: We classify countries into five quintiles based on their income level

in the 1950s (`quintile50s`, Q1 = poorest, Q5 = richest), a pre-determined variable that is orthogonal to subsequent democratization decisions. We then estimate the baseline dynamic panel specification of Acemoglu et al. (2019) — four lags of log GDP per capita, country and year fixed effects — separately for each quintile, using both the Within (FE) and 2SLS estimators. The instrument for democracy is the same regional democratization wave (`demreg`) as in the original paper, estimated with four lags. Splitting the sample by quintile rather than interacting democracy with quintile dummies avoids the perfect collinearity that arises when a time-invariant quintile assignment is interacted with democracy in a fixed-effects model.

Results

Table 11 reports the democracy coefficient and standard error for each quintile under both estimators. Figures 8 and 9 plot the implied democracy effect across quintiles with 95% confidence intervals.

Table 11: Replication of Acemoglu et al. (2019), Table 2 — Democracy and Growth by Initial Income Quintile (Extension 2)

	Within (FE)					2SLS (IV)				
	Q1 (Poorest)	Q2	Q3	Q4	Q5 (Richest)	Q1 (Poorest)	Q2	Q3	Q4	Q5 (Richest)
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
Democracy	0.286	0.958	0.338	0.238	1.513	-2.599	0.531	-0.556	1.859	0.744
	(0.452)	(0.409)	(0.529)	(0.780)	(0.786)	(2.650)	(2.650)	(1.502)	(1.194)	(1.693)
Observations	1102	1267	1071	911	1172	1102	1267	1044	911	1172

Notes: Replication of the estimation framework of Acemoglu et al. (2019), Table 2, extended to examine heterogeneity by initial income. Each column reports the democracy coefficient from a separate dynamic panel regression estimated on the subsample of countries in that initial income quintile. Quintiles are defined by 1950s GDP per capita (`quintile50s`), with Q1 the poorest and Q5 the richest. All specifications include four lags of log GDP per capita and country and year fixed effects. Columns (1)–(5) report Within (FE) estimates with standard errors clustered at the country level. Columns (6)–(10) report 2SLS estimates instrumenting democracy with four lags of the regional democratization wave (`demreg`); standard errors are clustered at the country level. The democracy coefficient is multiplied by 100. The first-stage F-statistic is below the conventional threshold of 10 for Q2 and Q3 in the IV panel, so those estimates should be interpreted with caution. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

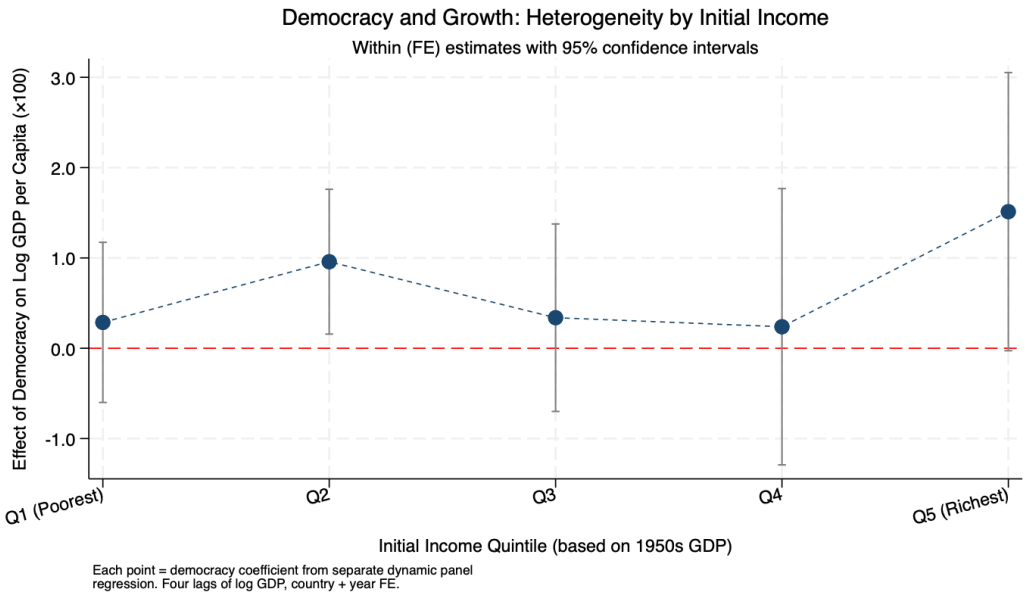


Figure 8: Replication of Acemoglu et al. (2019) (Table 2, Within/FE specification), extended by initial income quintile — Within (FE) estimates. Each point reports the democracy coefficient from a separate dynamic panel regression for countries in that income quintile, following the baseline specification of the original paper (four lags of log GDP per capita, country and year FE, clustered SEs). Bars are 95% confidence intervals. Q1 = poorest, Q5 = richest, classified by 1950s GDP per capita.

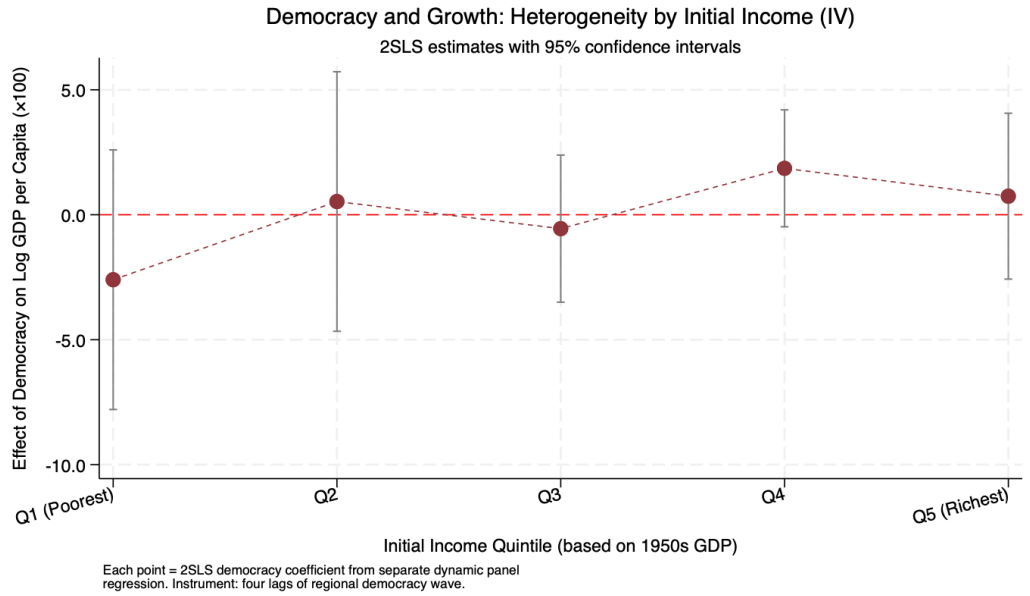


Figure 9: Replication of Acemoglu et al. (2019) (Table 2, 2SLS specification), extended by initial income quintile — 2SLS (IV) estimates. Each point reports the democracy coefficient from a separate dynamic panel 2SLS regression, instrumenting democracy with four lags of the regional democratization wave (`demreg`), following the original paper. Bars are 95% confidence intervals. Q1 = poorest, Q5 = richest. Note: first-stage F-statistics below 10 for Q2 and Q3; interpret those estimates with caution.

Interpretation

The results reveal a pattern that is neither uniformly pro-poor nor pro-rich, and that differs meaningfully between the FE and IV estimates.

Under the Within estimator (Figure 8), the democracy coefficient is positive across all five quintiles, consistent with the main finding of Acemoglu et al. (2019). The effect is smallest for Q4 (0.238) and largest for Q5 (1.513, significant at 10%) and Q2 (0.958, significant at 5%). This U-shaped pattern suggests that the FE democracy–growth relationship is actually strongest at the extremes of the income distribution, with the weakest effect in the upper-middle income range. However, only Q2 and Q5 achieve conventional significance levels, so the within-quintile variation should be interpreted cautiously.

The IV estimates (Figure 9) tell a more nuanced story. The democracy coefficient is negative for Q1 (-2.599, SE $= 2.650$) and Q3 (-0.556, SE $= 1.502$), and positive for Q2, Q4, and Q5, though none of the estimates are statistically significant. The wide confidence intervals in the IV panel reflect the smaller within-quintile sample sizes (roughly 1,000–1,300 observations each) and, for Q2 and Q3, weak first-stage F-statistics (3.46 and 1.98 respectively), suggesting that the regional democratization wave instrument has limited predictive power within those subsamples. The IV estimates for Q1 and Q4 are more credible (first-stage F-stats of 8.27 and 15.12), and the pattern — negative for the poorest, positive for upper-middle income — is consistent with the hypothesis that institutional capacity moderates the growth payoff to democratization.

Taken together, these findings suggest two conclusions. First, the positive aggregate democracy–growth effect documented by Acemoglu et al. (2019) is not uniformly distributed across the income spectrum: the causal (IV) estimates are most positive for Q4 countries, consistent with the view that a minimum level of institutional and economic development is needed to translate political liberalization into income gains. Second, the negative (though imprecise) IV estimate for Q1 points to a potential *capacity trap*: the poorest democratizing countries may face a short-run growth cost if democratic transitions divert resources toward political accommodation without the institutional infrastructure to sustain productive investment. A natural direction for future work is to interact the quintile subgroup analysis with measures of state capacity or rule of law to test this mechanism more directly.

5. Conclusion

This paper replicates the main-text results of Acemoglu et al. (2019) and develops two extensions: one testing whether democracy and market reforms are complementary drivers of growth, and one examining whether the aggregate democracy–growth effect is uniform across the initial income distribution.

Our replication closely matches the published findings across all eight tables and six figures. All deterministic estimators — within fixed-effects and system GMM — reproduce the original coefficients exactly. The only discrepancies arise in the bootstrap-based minimum distance (HHK) panels, where differences of a few tenths of a percentage point are consistent with simulation variance across random seeds rather than methodological failures. In Table 6, minor deviations in the spatial-lag columns (8)–(9) reflect the unavailability of the original `spmat` package and are addressed using a `geodist`-based alternative that yields qualitatively identical results. We also

document and correct two bugs in the original Table 8 do-file: a misspecified instrument for the schooling interaction term, and a unit-of-measurement inconsistency in the schooling variables. These corrections bring the 2SLS heterogeneous effects estimates into exact agreement with the published table.

Our extension provides partial evidence that the growth return to democratization is larger in countries with more market-oriented institutions. The interaction between democracy and the market reforms index is positive and statistically significant once the direct effect of reform levels is absorbed, and the democracy–growth effect is strongly concentrated in the post-Cold War period, consistent with the simultaneous wave of democratization and liberalization in the 1990s. These findings suggest that the economic benefits of democracy may be context-dependent, and that institutional complementarities between political and economic liberalization deserve further investigation.

Our second extension examines whether the aggregate democracy–growth effect masks heterogeneity across the initial income distribution. Splitting the sample into five quintiles based on 1950s GDP per capita and estimating the baseline specification separately for each, we find that the Within (FE) estimates are positive across all quintiles — consistent with the main result — but that the IV estimates reveal a more nuanced picture. The causal effect is most positive for upper-middle income countries (Q4), while the estimate for the poorest quintile (Q1) is negative, though imprecisely estimated. This pattern is consistent with a capacity trap hypothesis: countries at the very bottom of the income distribution may lack the institutional infrastructure to translate democratic transitions into sustained growth, at least in the short run.

A key limitation of the extensions is identification: in Extension 1 the interaction term is not separately instrumented, and in Extension 2 the within-quintile IV estimates suffer from weak instruments for some subgroups. An important direction for future work is to find credible exogenous variation in market reforms — for instance from IMF conditionality or multilateral trade agreements — and to interact the quintile analysis with direct measures of state capacity to sharpen the distributional conclusions.

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A. Original Tables and Figures from Acemoglu et al. (2019)

This appendix reproduces the original tables and figures from Acemoglu et al. (2019) that correspond to the replication results presented in the main text. The appendix is provided for reference so that readers can directly compare the replicated estimates with the published results.

A.1. Original Figures

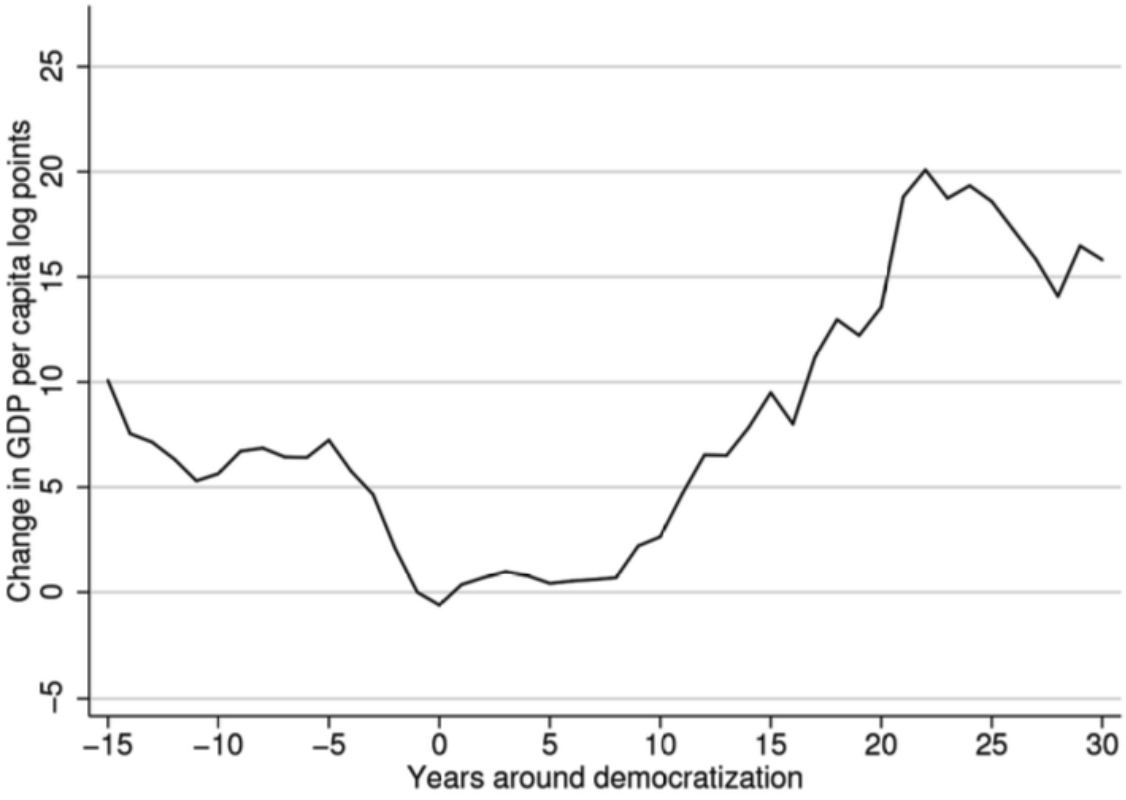


FIG. 1.—GDP per capita before and after a democratization. This figure plots GDP per capita in log points around a democratic transition relative to countries remaining non-democratic in the same year. We normalize log GDP per capita to 0 in the year preceding the democratization. Time (in years) relative to the year of democratization runs on the horizontal axis.

Figure 10: Original Figure 1: GDP per capita before and after democratization

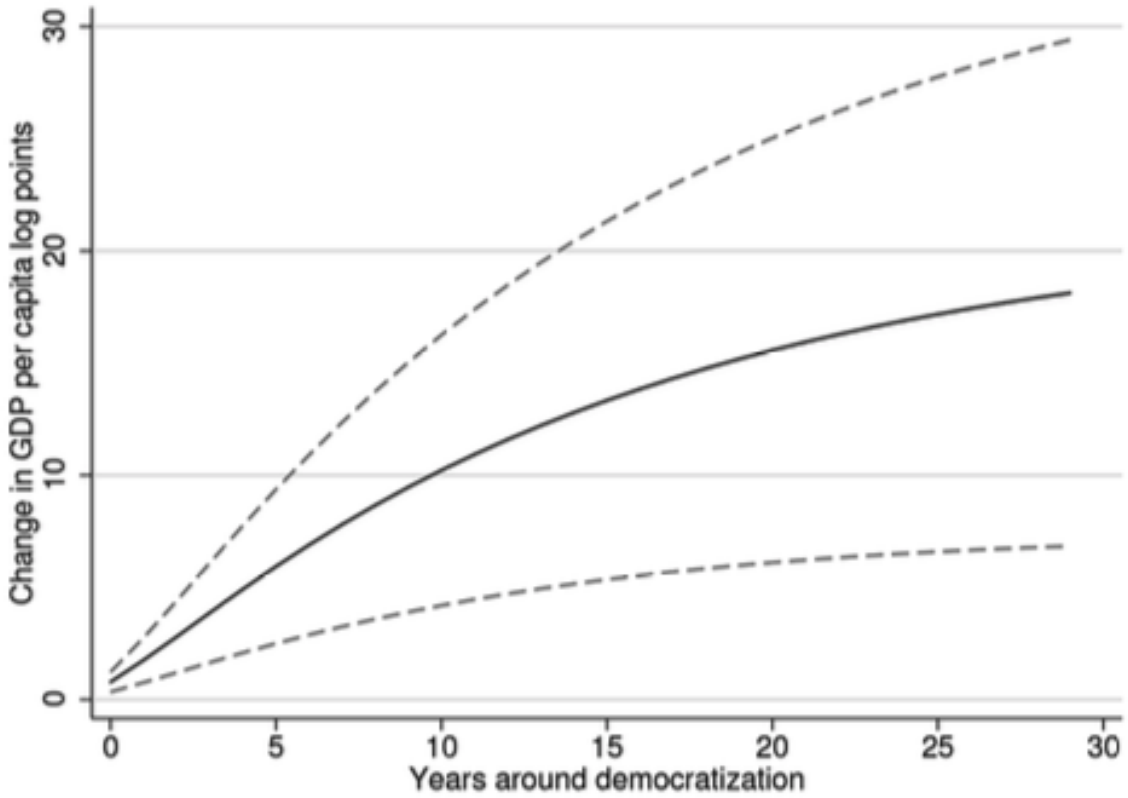


Figure 11: Original Figure 2: Dynamic panel estimates of the effect of democracy

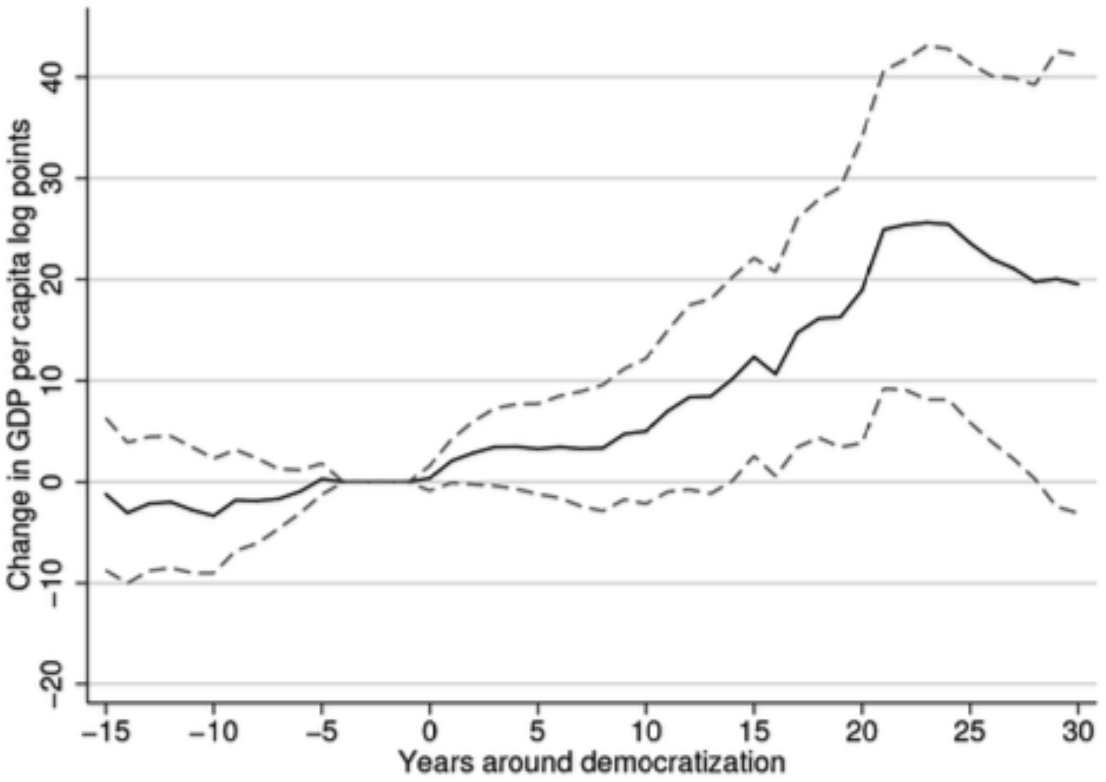


FIG. 3.—Semiparametric estimates of the over-time effects of democracy on the log of GDP, obtained with a regression model to estimate counterfactuals. This figure plots semiparametric estimates of the effect of democratization on GDP per capita in log points. The solid line plots the estimated average effect on GDP per capita on countries that democratized (in log points), with a 95 percent confidence interval in dashed lines. Time (in years) relative to the year of democratization runs on the horizontal axis. The estimates are obtained by assuming and estimating a linear model for counterfactual outcomes, which we use to control for the influence of GDP dynamics. Section IV explains our approach in full detail.

Figure 12: Original Figure 3: Semiparametric estimates using regression adjustment

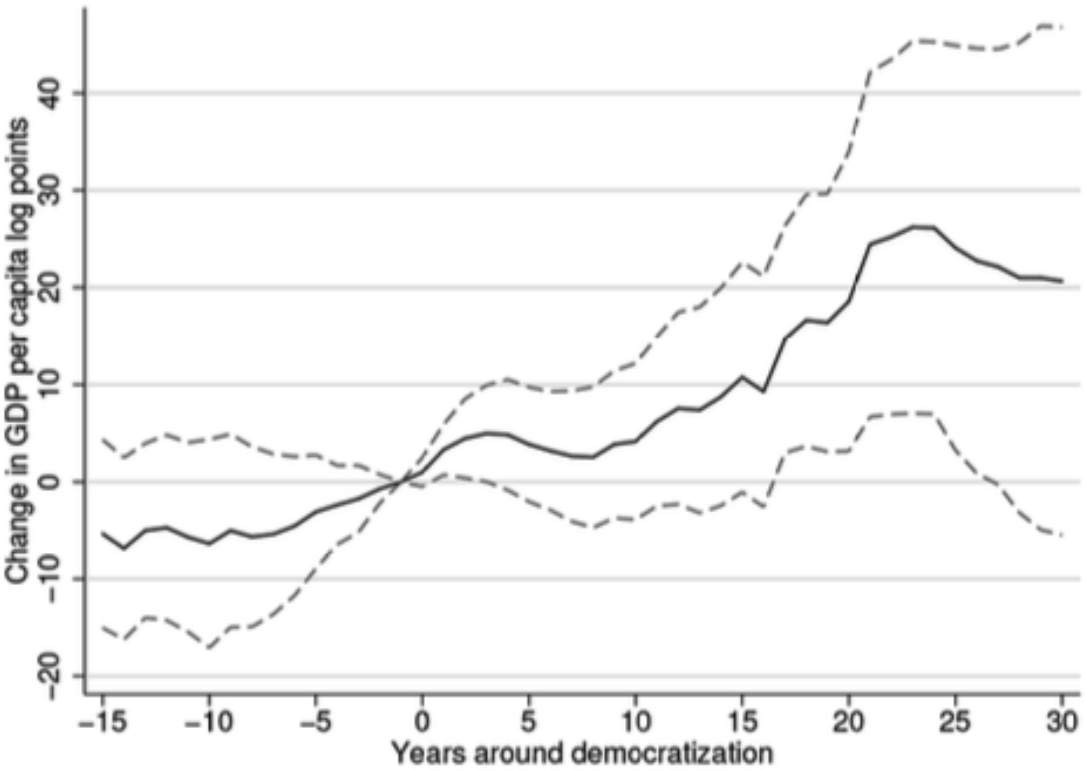


FIG. 4.—Semiparametric estimates of the over-time effects of democracy on the log of GDP, obtained with inverse-propensity-score reweighting. This figure plots semiparametric estimates of the effect of democratizations on GDP per capita in log points. The solid line plots the estimated average effect on GDP per capita on countries that democratized (in log points), with a 95 percent confidence interval in dashed lines. Time (in years) relative to the year of democratization runs on the horizontal axis. The estimates are obtained by assuming and estimating a probit model for democratizations based on GDP lags, which we use to estimate the propensity score and reweight the data. Section IV explains our approach in full detail.

Figure 13: Original Figure 4: Semiparametric estimates using IPW

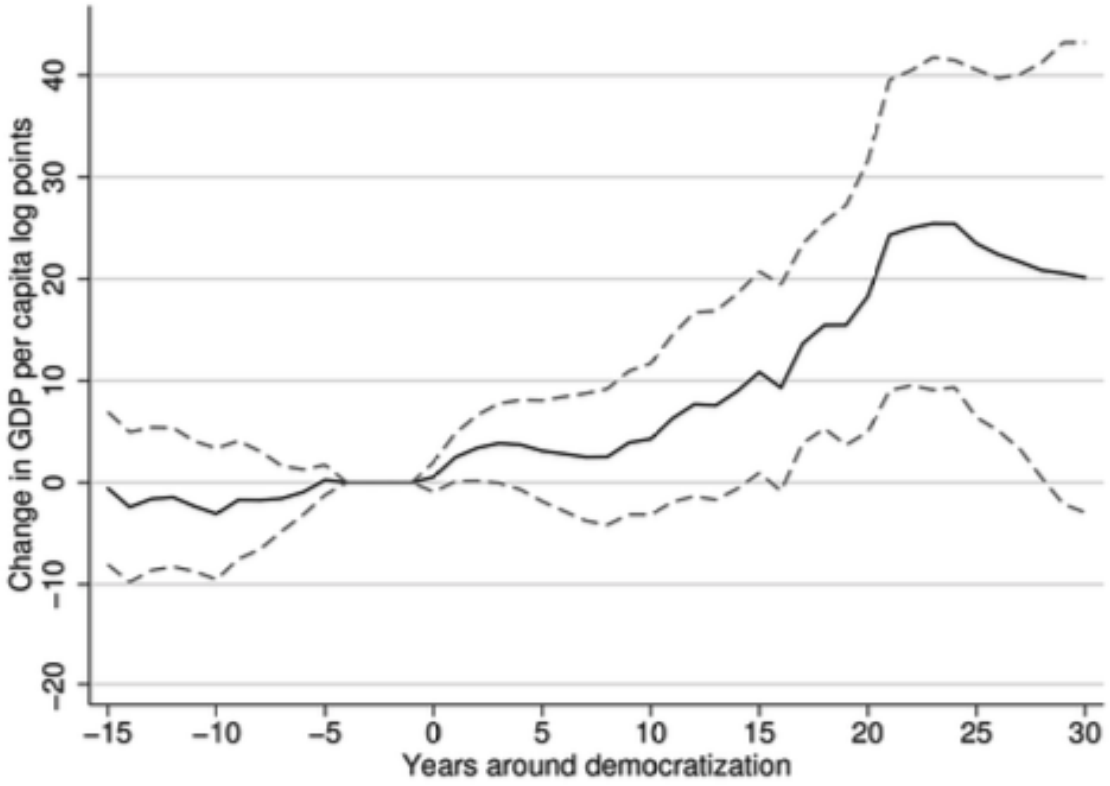


FIG. 5.—Semiparametric estimates of the over-time effects of democracy on the log of GDP; doubly robust estimates. This figure plots semiparametric estimates of the effect of democratization on GDP per capita in log points. The solid line plots the estimated average effect on GDP per capita on countries that democratized (in log points), with a 95 percent confidence interval in dashed lines. Time (in years) relative to the year of democratization runs on the horizontal axis. The estimates are obtained by assuming and estimating a probit model for democratizations based on GDP lags, which we use to estimate the propensity score and reweight the data. In addition, we partial out lags of GDP linearly, making our approach doubly robust. Section IV explains our approach in full detail.

Figure 14: Original Figure 5: Doubly robust semiparametric estimates

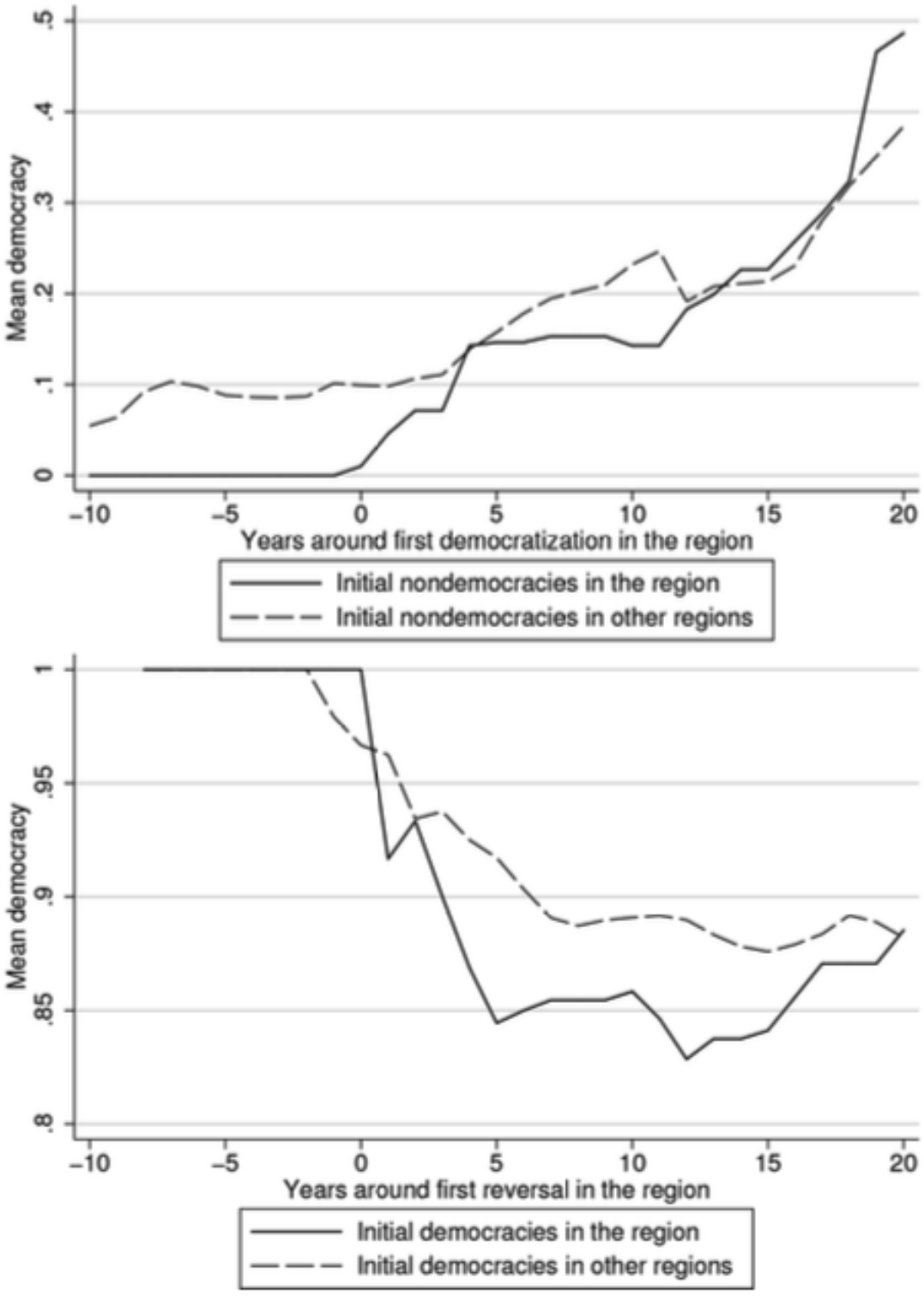


FIG. 6.—Regional democratizations and reversal waves. These figures illustrate the existence of regional democracy waves. The top figure plots average democracy among initial nondemocracies around the first democratization in the region. For comparison, it also plots average democracy among other initial nondemocracies in other regions. The bottom figure plots average democracy among initial democracies around the first reversal in the region. For comparison, it also plots average democracy among other initial democracies in other regions.

Figure 15: Original Figure 6: Regional democratizations and reversal waves

A.2. Original Tables

TABLE 1
Summary Statistics for the Main Variables Used in Our Analysis

	Nondemocracies			Democracies		
	Observations	Mean	Standard Deviation	Observations	Mean	Standard Deviation
GDP per capita	3,376	$2,074	$3,838	3,558	$8,149	$9,334
Investment share of GDP	3,012	.22	.10	3,300	.23	.07
Trade share of GDP	2,959	.71	.50	3,427	.77	.41
Primary-school enrollment rate	2,305	90.13%	27.47%	2,771	101.60%	15.72%
Secondary-school enrollment rate	1,970	44.76%	31.09%	2,500	75.78%	29.62%
Tax revenue share of GDP	2,524	.15	.08	2,442	.20	.09
Child mortality per 1,000 births	3,185	76.75	47.58	3,492	33.07	32.87
Unrest rate	2,911	29.44%	45.58%	3,419	22.57%	41.81%
Market reforms index (0–100)	2,533	26.89	23.09	2,711	52.73	24.61

Note.—See the text for a full description of the variables and their corresponding sources. The table presents the statistics separately for nondemocracies (countries/years for which our dichotomous democracy measure is 0) and democracies (countries/years for which our dichotomous democracy measure is 1).

Figure 16: Original Table 1: Summary statistics

TABLE 2
Effect of Democracy on (Log) GDP per Capita

	Within Estimates				Arellano and Bond Estimates				HHK Estimates			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Democracy	.973	.651	.787	.887	.959	.797	.875	.659	.781	.582	1.178	1.682
	(.294)	(.248)	(.226)	(.245)	(.477)	(.417)	(.374)	(.378)	(.455)	(.387)	(.370)	(.352)
Log GDP, first lag	.973	1.266	1.238	1.233	.946	1.216	1.204	1.204	.938	1.158	1.150	1.155
	(.006)	(.038)	(.038)	(.039)	(.009)	(.041)	(.041)	(.038)	(.011)	(.038)	(.040)	(.036)
Log GDP, second lag		−.300	−.207	−.214		−.270	−.193	−.205		−.217	−.127	−.122
		(.037)	(.046)	(.043)		(.038)	(.045)	(.042)		(.035)	(.050)	(.041)
Log GDP, third lag			−.026	−.021			−.028	−.020			−.030	−.040
			(.028)	(.028)			(.028)	(.027)			(.026)	(.024)
Log GDP, fourth lag			−.043	−.039			−.036	−.038			−.039	−.028
			(.017)	(.034)			(.020)	(.033)			(.015)	(.026)
p-value, lags 5–8				.565				.478				.094
Long-run effect of democracy	35.587	19.599	21.240	22.008	17.608	14.882	16.448	11.810	12.644	9.929	25.032	35.104
	(13.998)	(8.595)	(7.215)	(7.740)	(10.609)	(9.152)	(8.436)	(7.829)	(8.282)	(7.258)	(10.581)	(11.140)
Effect of democracy after 25 years	17.791	13.800	16.895	17.715	13.263	12.721	14.713	10.500	10.076	8.537	20.853	29.528
	(5.649)	(5.550)	(5.297)	(5.455)	(7.281)	(7.371)	(7.128)	(6.653)	(6.245)	(6.032)	(7.731)	(7.772)
Persistence of GDP process	.973	.967	.963	.960	.946	.946	.947	.944	.938	.941	.953	.952
	(.006)	(.005)	(.005)	(.007)	(.009)	(.009)	(.009)	(.009)	(.011)	(.010)	(.009)	(.009)
AR2 test *p*-value					.01	.08	.51	.95				
Unit root test *t*-statistics	−4.79	−3.89	−4.13	−7.00								
p-value (reject unit root)	.00	.00	.00	.00								
Observations	6,790	6,642	6,336	5,688	6,615	6,467	6,161	5,513	6,615	6,467	6,161	5,513
Countries in sample	175	175	175	175	175	175	175	175	175	175	175	175

Note.—This table presents estimates of the effect of democracy on log GDP per capita. The reported coefficient on democracy is multiplied by 100. Cols. 1–4 present results from the within estimator. Cols. 5–8 present results from Arellano and Bond's (1991) GMM estimator. The AR2 row reports the *p*-value for a test of serial correlation in the residuals of the GDP series. Cols. 9–12 present results from the HHK (Hahn et al. 2001) estimator. In all specifications we control for a full set of country and year fixed effects. Cols. 4, 8, and 12 include eight lags of GDP per capita as controls, but we report only the *p*-value of a test for joint significance of lags 5–8. Standard errors robust against heteroskedasticity and serial correlation at the country level are reported in parentheses.

Figure 17: Original Table 2: Effect of democracy on log GDP per capita

TABLE 3
EFFECT OF DEMOCRACY ON THE GROWTH RATE OF GDP PER CAPITA

	Within Estimates				Arellano and Bond Estimates				HHK Estimates			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Democracy	1.028	1.201	1.269	1.378	1.458	1.715	1.545	1.554	1.410	1.413	1.343	1.253
	(.250)	(.237)	(.243)	(.284)	(.385)	(.385)	(.368)	(.405)	(.323)	(.341)	(.348)	(.330)
GDP growth, first lag	.287	.274	.263	.250	.294	.278	.270	.254	.245	.230	.211	.211
	(.037)	(.037)	(.039)	(.037)	(.040)	(.038)	(.041)	(.036)	(.034)	(.037)	(.039)	(.036)
GDP growth, second lag		.047	.060	.042		.057	.064	.047		.075	.082	.057
		(.024)	(.025)	(.023)		(.025)	(.025)	(.024)		(.023)	(.025)	(.020)
GDP growth, third lag			.023	.024			.031	.028			.029	.036
			(.018)	(.017)			(.018)	(.018)			(.015)	(.021)
GDP growth, fourth lag			−.033	−.018			−.022	−.013			−.007	−.004
			(.022)	(.025)			(.022)	(.025)			(.015)	(.020)
p-value, lags 5–8				.00				.00				.02
Long-run effect of democracy (on growth rate)	1.443	1.768	1.845	1.960	2.065	2.579	2.349	2.273	1.867	2.032	1.961	1.790
	(.360)	(.372)	(.370)	(.441)	(.559)	(.624)	(.602)	(.644)	(.448)	(.511)	(.527)	(.488)
Level effect of democracy after 25 years	35.483	43.245	45.275	42.715	50.767	62.957	57.296	50.090	46.067	49.685	47.791	40.543
	(8.820)	(9.046)	(9.038)	(9.191)	(13.718)	(15.105)	(14.572)	(13.322)	(11.017)	(12.448)	(12.770)	(10.837)
Persistence of growth rate process	.287	.321	.312	.297	.294	.335	.342	.316	.245	.305	.315	.300
	(.037)	(.041)	(.038)	(.043)	(.040)	(.043)	(.041)	(.044)	(.034)	(.039)	(.043)	(.049)
AR2 test *p*-value					.10	.81	.98	.93				
Observations	6,642	6,490	6,178	5,523	6,467	6,315	6,003	5,348	6,467	6,315	6,003	5,348
Countries in sample	175	175	175	175	175	175	175	174	175	175	175	174

NOTE.—This table presents estimates of the effect of democracy on the growth rate of GDP per capita. The reported coefficient on democracy is multiplied by 100. Cols. 1–4 present results from the within estimator. Cols. 5–8 present results from Arellano and Bond's (1991) GMM estimator. The AR2 row reports the *p*-value for a test of serial correlation in the residuals of the growth rate series. Cols. 9–12 present results from the HHK (Hahn et al. 2001) estimator. In all specifications we control for a full set of country and year fixed effects. Cols. 4, 8, and 12 include eight lags of the growth rate of GDP per capita as controls, but we report only the *p*-value of a test for joint significance of lags 5–8. Standard errors robust against heteroskedasticity and serial correlation at the country level are reported in parentheses.

Figure 18: Original Table 3: Effect of democracy on GDP growth

TABLE 4
EFFECT OF DEMOCRACY ON (LOG) GDP PER CAPITA, CONTROLLING FOR COVARIATES

		COVARIATES INCLUDED						
	(1)	GDP in 1960 Quintiles × Year Effects (2)	Soviet Dummies (3)	Lags of Unrest (4)	Lags of Trade (5)	Lags of Financial Flows (6)	Lags of Demographic Structure (7)	Region × Regime × Year Effects (8)
				A. Within Estimates				
Democracy	.787	.718	.911	.705	.595	.926	.650	.834
	(.226)	(.249)	(.251)	(.224)	(.264)	(.244)	(.230)	(.264)
Long-run effect of democracy	21.240	22.173	24.860	17.000	14.593	23.870	14.153	16.651
	(7.215)	(8.702)	(7.783)	(5.980)	(7.122)	(8.211)	(5.419)	(5.546)
Effect of democracy after 25 years	16.895	16.261	19.587	13.567	11.500	18.149	12.251	14.532
	(5.297)	(5.982)	(5.724)	(4.644)	(5.336)	(5.435)	(4.552)	(4.726)
Persistence of GDP process	.963	.968	.963	.959	.959	.961	.954	.950
	(.005)	(.005)	(.005)	(.004)	(.006)	(.006)	(.005)	(.005)
Observations	6,336	5,523	6,336	5,643	5,750	4,950	6,262	6,336
Countries in sample	175	149	175	171	172	171	172	175
				B. Arellano and Bond Estimates				
Democracy	.875	.730	1.073	.693	1.034	1.017	.756	1.217
	(.374)	(.387)	(.403)	(.396)	(.469)	(.373)	(.370)	(.420)
Long-run effect of democracy	16.448	14.865	20.006	9.871	17.926	18.607	12.152	18.209
	(8.436)	(8.998)	(8.981)	(6.479)	(9.021)	(7.842)	(6.639)	(6.746)
Effect of democracy after 25 years	14.713	12.759	17.874	9.159	15.659	15.903	11.334	16.861
	(7.128)	(7.350)	(7.564)	(5.768)	(7.593)	(6.327)	(6.004)	(6.050)
Persistence of GDP process	.947	.951	.946	.930	.942	.945	.938	.933
	(.009)	(.008)	(.009)	(.012)	(.009)	(.007)	(.010)	(.010)
AR2 test p-value	.51	.90	.28	.62	.72	.34	.58	.70
Observations	6,161	5,374	6,161	5,467	5,570	4,779	6,090	6,161
Countries in sample	175	149	175	171	172	171	172	175
				C. HHK Estimates				
Democracy	1.178	.722	1.059	1.203	1.110	2.030	1.262	1.482
	(.370)	(.357)	(.364)	(.376)	(.332)	(.359)	(.355)	(.449)
Long-run effect of democracy	25.032	15.731	21.648	25.557	24.575	32.631	22.161	26.358
	(10.581)	(8.476)	(9.431)	(9.842)	(9.031)	(7.727)	(6.641)	(9.178)
Effect of democracy after 25 years	20.853	12.719	18.313	20.753	19.407	28.896	19.633	22.776
	(7.731)	(6.503)	(7.162)	(7.072)	(6.359)	(6.223)	(5.647)	(7.380)
Persistence of GDP process	.953	.954	.951	.953	.955	.938	.943	.944
	(.009)	(.006)	(.009)	(.008)	(.008)	(.008)	(.006)	(.008)
Observations	6,161	5,374	6,161	5,467	5,570	4,779	6,090	6,161
Countries in sample	175	149	175	171	172	171	172	175

NOTE.—This table presents estimates of the effect of democracy on log GDP per capita. The reported coefficient of democracy is multiplied by 100. Panel A presents results from the within estimator. Panel B presents results from Arellano and Bond's GMM (1991) estimator. The AR2 row reports the p-value for a test of serial correlation in the residuals of the GDP series. Panel C presents results using the HHK (Hahn et al. 2001) estimator. In all specifications we control for a full set of country and year fixed effects and four lags of GDP per capita. In addition, we control for the covariates specified in each column label and described in the text. Standard errors robust against heteroskedasticity and serial correlation at the country level are reported in parentheses.

Figure 19: Original Table 4: Effect of democracy controlling for covariates

TABLE 5
SEMIPARAMETRIC ESTIMATES OF THE EFFECT OF DEMOCRATIZATIONS ON (LOG) GDP PER CAPITA

	AVERAGE EFFECTS FROM						
	−5 to −1 Years (1)	0–4 Years (2)	5–9 Years (3)	10–14 Years (4)	15–19 Years (5)	20–24 Years (6)	25–29 Years (7)
	A. Linear Regression Adjustment						
Avg. effect on log GDP	.060	2.454	3.621	7.806	14.037	24.075	21.310
	(.156)	(1.382)	(2.792)	(4.416)	(5.384)	(8.262)	(9.643)
	B. Inverse-Propensity-Score Reweighting						
Avg. effect on log GDP	−1.586	3.724	3.214	6.818	13.542	24.111	22.184
	(1.478)	(1.789)	(3.327)	(4.848)	(5.892)	(9.035)	(11.561)
	C. Doubly Robust Estimator						
Avg. effect on log GDP	.051	2.795	2.969	6.966	12.947	23.691	21.793
	(.151)	(1.471)	(3.067)	(4.359)	(4.881)	(7.638)	(9.566)

NOTE.—This table presents semiparametric estimates of the effect of a democratization on log GDP per capita over different time horizons, indicated in the column labels. We report estimates of the average (avg.) effect on the treated. Panel A presents estimates using regression adjustment to compute counterfactual outcomes for treated countries. Panel B presents estimates obtained via inverse-propensity-score reweighting. Panel C presents estimates obtained with a doubly robust estimator, combining the regression adjustment and the inverse-propensity-score reweighting. Below each estimate we report robust standard errors obtained via bootstrapping.

Figure 20: Original Table 5: Semiparametric estimates

TABLE 6
IV Estimates of the Effect of Democracy on (Log) GDP per Capita

			Covariates Included						
	(1)	(2)	GDP in 1960 Quintiles × Year Effects (3)	Soviet Dummies (4)	Regional Trends (5)	Regional GDP and Trade (6)	Regional Unrest GDP and Trade (7)	Spatial Lag of GDP (8)	Spatial Lags of GDP and Democracy (9)
				A. 2SLS Estimates with Fixed Effects					
Democracy	.966	1.149	1.125	1.292	1.697	1.817	1.107	1.335	1.361
	(.558)	(.554)	(.689)	(.651)	(.885)	(.663)	(.656)	(.536)	(.895)
Long-run effect of democracy	26.315	31.521	35.226	35.723	36.788	41.544	25.016	37.482	38.439
	(17.075)	(17.425)	(23.846)	(19.997)	(20.657)	(17.157)	(16.002)	(17.836)	(27.883)
Effect of democracy after 25 years	20.836	24.866	25.618	27.929	32.051	35.350	21.386	29.217	29.011
	(12.862)	(12.978)	(16.538)	(14.944)	(17.703)	(14.017)	(13.342)	(12.894)	(19.692)
Persistence of GDP process	.963	.964	.968	.964	.954	.956	.956	.964	.965
	(.005)	(.005)	(.005)	(.005)	(.006)	(.006)	(.006)	(.005)	(.006)
Hansen p-value		.21	.18	.32	.28	.25	.09	.04	.19
Observations	6,312	6,309	5,496	6,309	6,309	6,309	6,309	6,181	6,009
Countries in sample	174	174	148	174	174	174	174	173	173
Excl.-instruments Fstatistic	119.1	33.2	16.8	26.7	23.7	13.6	16.7	17.5	4.6
				B. First-Stage Estimates					
Democracy wave $t-1$	.800	.547	.503	.480	.498	.522	.508	.540	.586
	(.073)	(.101)	(.130)	(.099)	(.092)	(.104)	(.102)	(.103)	(.101)
Democracy wave $t-2$		.133	.109	.133	.129	.117	.115	.136	.128
		(.081)	(.094)	(.080)	(.081)	(.079)	(.078)	(.078)	(.088)
Democracy wave $t-3$		.227	.270	.223	.228	.221	.223	.224	.282
		(.067)	(.077)	(.065)	(.070)	(.069)	(.070)	(.070)	(.077)
Democracy wave $t-4$		−.087	−.119	−.075	−.123	−.083	−.064	−.072	−.107
		(.110)	(.126)	(.110)	(.106)	(.113)	(.113)	(.113)	(.116)
				C. HHK Estimates					
Democracy	.690	.944	1.435	.719	.822	1.311	.897	1.021	1.206
	(.642)	(.479)	(.599)	(.503)	(.480)	(.435)	(.371)	(.549)	(.485)
Long-run effect of democracy	14.512	24.766	46.767	18.337	16.413	24.040	17.290	29.286	31.111
	(14.703)	(14.083)	(22.556)	(13.688)	(10.700)	(9.989)	(8.556)	(18.354)	(15.167)
Effect of democracy after 25 years	11.768	18.670	31.039	13.969	13.778	21.100	14.668	21.133	23.702
	(11.445)	(9.799)	(13.113)	(9.935)	(8.523)	(8.038)	(6.734)	(11.942)	(10.243)
Persistence of GDP process	.952	.962	.969	.961	.950	.945	.948	.965	.961
	(.011)	(.008)	(.008)	(.009)	(.010)	(.010)	(.010)	(.009)	(.008)
Observations	6,161	6,161	5,374	6,161	6,161	6,161	6,161	6,132	5,960
Countries in sample	174	174	148	174	174	174	174	173	173

Note.—This table presents IV estimates of the effect of democracy on log GDP per capita. The reported coefficient of democracy is multiplied by 100. Panel A presents 2SLS estimates instrumenting democracy with up to four lags of regional democracy waves and the p-value of a Hansen overidentification test. Panel B presents the corresponding first-stage estimates and the excluded (excl.)-instruments F statistic. Panel C presents results using the HHK (Hahn et al. 2001) estimator instrumenting democracy with up to four lags of regional democracy waves (except for col. 1, where we use only one lag). In all specifications we control for a full set of country and year fixed effects and four lags of GDP per capita. In addition, we control for the covariates specified in each column label and described in the text. Standard errors robust against heteroskedasticity and serial correlation at the country level are in parentheses.

Figure 21: Original Table 6: IV estimates

TABLE 7
EFFECTS OF DEMOCRACY ON POTENTIAL MECHANISMS

	Dependent Variable								
	Log of Investment Share in GDP (1)	Log of TFP (2)	Index of Economic Reforms (3)	Log of Trade Share in GDP (4)	Log of Tax Share in GDP (5)	Log of Primary-School Enrollment (6)	Log of Secondary-School Enrollment (7)	Log of Child Mortality (8)	Dummy for Unrest (9)
	A. Within Estimates								
Democracy	2.391	−.205	.687	.689	3.311	1.042	1.345	−.253	−7.832
	(1.114)	(.276)	(.348)	(.676)	(1.409)	(.338)	(.610)	(.063)	(2.185)
Long-run effect of democracy	9.112	−2.883	5.580	5.445	16.062	21.908	18.960	−34.264	−11.944
	(4.255)	(3.858)	(2.883)	(5.253)	(6.650)	(7.624)	(8.622)	(10.747)	(3.329)
Effect of democracy after 25 years	9.089	−2.738	5.359	5.303	15.864	18.892	18.057	−21.400	−11.944
	(4.245)	(3.648)	(2.753)	(5.126)	(6.574)	(6.321)	(8.146)	(5.124)	(3.329)
Persistence of outcome process	.738	.929	.877	.873	.794	.952	.929	.993	.344
	(.020)	(.012)	(.012)	(.011)	(.016)	(.008)	(.013)	(.001)	(.030)
Observations	5,665	3,879	4,692	5,738	4,511	3,714	2,883	6,084	5,646
Countries in sample	169	107	150	172	131	166	158	173	171
	B. 2SLS Estimates								
Democracy	2.211	−.941	3.224	5.512	8.088	1.757	4.116	−.715	−5.569
	(2.852)	(.667)	(.863)	(2.005)	(3.021)	(.721)	(1.626)	(.164)	(5.682)
Long-run effect of democracy	8.440	−12.738	23.775	40.589	38.609	36.693	57.072	−95.728	−8.471
	(10.705)	(8.854)	(6.215)	(13.580)	(14.330)	(15.505)	(21.698)	(26.347)	(8.577)
Effect of democracy after 25 years	8.419	−12.167	23.156	39.817	38.159	31.611	54.252	−58.625	−8.471
	(10.681)	(8.380)	(6.039)	(13.375)	(14.121)	(12.863)	(20.267)	(13.123)	(8.577)
Persistence of outcome process	.738	.926	.864	.864	.791	.952	.928	.993	.343
	(.020)	(.012)	(.012)	(.012)	(.017)	(.008)	(.013)	(.001)	(.030)
Excl.-instruments *F*-statistic	21.7	27.7	43.7	21.5	31.8	12.1	10.4	26.3	28.6
Hansen *p*-value	.29	.06	.22	.09	.69	.09	.12	.02	.84
Observations	5,640	3,871	4,670	5,714	4,489	3,710	2,879	6,057	5,619
Countries in sample	168	107	149	171	130	164	156	172	170

TABLE 7 (*Continued*)

	Dependent Variable								
	Log of Investment Share in GDP (1)	Log of TFP (2)	Index of Economic Reforms (3)	Log of Trade Share in GDP (4)	Log of Tax Share in GDP (5)	Log of Primary-School Enrollment (6)	Log of Secondary-School Enrollment (7)	Log of Child Mortality (8)	Dummy for Unrest (9)
	C. HHK Estimates								
Democracy	6.603	.388	1.121	1.255	4.277	1.384	2.144	−.306	−3.638
	(1.336)	(.294)	(.371)	(.790)	(2.044)	(.366)	(.644)	(.068)	(2.931)
Long-run effect of democracy	25.495	7.518	22.655	10.182	24.622	41.349	43.070	−54.798	−5.742
	(5.313)	(6.011)	(11.199)	(6.584)	(11.858)	(14.855)	(15.445)	(15.745)	(4.630)
Effect of democracy after 25 years	25.432	6.748	15.698	9.807	23.966	29.049	36.865	−29.139	−5.742
	(5.294)	(5.366)	(5.953)	(6.307)	(11.461)	(8.614)	(11.888)	(6.131)	(4.630)
Persistence of outcome process	.741	.948	.951	.877	.826	.967	.950	.994	.366
	(.018)	(.009)	(.018)	(.014)	(.031)	(.007)	(.012)	(.001)	(.037)
Observations	5,125	3,557	4,236	4,866	4,045	3,579	2,683	5,454	5,233
Countries in sample	168	107	149	171	130	164	156	172	170

NOTE.—This table presents estimates of the effect of democracy on the different channels specified in the columns labels. The reported coefficient of democracy is multiplied by 100 (except for cols. 3 and 9). Panel A presents within estimates. Panel B presents 2SLS estimates instrumenting democracy with four lags of regional democracy waves, the *F*-statistic for the excluded (excl.) instruments, and the *p*-value of Hansen's overidentification test. Panel C presents results using the HHK (Hahn et al. 2001) estimator instrumenting democracy with four lags of regional democracy. In all specifications we control for a full set of country and year fixed effects, four lags of GDP per capita, and four lags of the dependent variable. Standard errors robust against heteroskedasticity and serial correlation at the country level are in parentheses.

Figure 22: Original Table 7: Mechanisms

TABLE 8
HETEROGENEOUS EFFECTS OF DEMOCRACY ON (LOG) GDP PER CAPITA

	Interaction with Log GDP per Capita				Interaction with Share with Secondary Education			
	1960 (1)	1970 (2)	1980 (3)	Lagged (4)	1960 (5)	1970 (6)	1980 (7)	Lagged (8)
	A. Within Estimates							
Democracy	.432	.572	.687	.744	.446	.340	.385	.495
	(.275)	(.248)	(.248)	(.246)	(.254)	(.253)	(.246)	(.241)
Interaction	.001	.001	.002	.001	.046	.049	.038	.020
	(.002)	(.001)	(.002)	(.002)	(.028)	(.020)	(.014)	(.013)
Long-run effect of democracy	16.231	18.631	20.489	19.843	13.785	10.480	11.841	14.597
	(11.160)	(9.073)	(8.608)	(8.255)	(8.550)	(8.275)	(8.118)	(8.432)
Effect of democracy after 25 years	10.013	12.916	14.985	15.877	10.081	7.679	8.687	10.953
	(6.565)	(5.960)	(5.848)	(5.943)	(5.964)	(5.872)	(5.728)	(5.821)
Persistence of GDP process	.973	.969	.966	.963	.968	.968	.967	.966
	(.005)	(.005)	(.005)	(.006)	(.005)	(.005)	(.005)	(.006)
Observations	4,281	4,909	5,525	6,336	5,300	5,300	5,300	5,300
Countries in sample	93	109	131	175	138	138	138	138
	B. 2SLS Estimates							
Democracy	.500	.155	.645	1.326	−.119	−.484	−.474	.600
	(1.088)	(.961)	(.929)	(.887)	(.662)	(.665)	(.639)	(.576)
Interaction	−.002	.000	−.000	−.003	.174	.156	.116	.049
	(.005)	(.004)	(.004)	(.004)	(.060)	(.047)	(.033)	(.023)
Long-run effect of democracy	18.838	4.978	19.275	36.116	−3.649	−14.586	−14.135	17.373
	(43.554)	(31.473)	(30.208)	(29.900)	(19.968)	(19.023)	(18.114)	(18.629)
Effect of democracy after 25 years	11.592	3.486	14.078	28.377	−2.692	−10.843	−10.574	13.133
	(25.784)	(21.795)	(21.085)	(21.317)	(14.837)	(14.524)	(13.901)	(13.312)
Persistence of GDP process	.973	.969	.967	.963	.967	.967	.966	.965
	(.006)	(.006)	(.006)	(.008)	(.006)	(.006)	(.006)	(.006)
Excl.-instruments *F*-statistic	6.6	6.1	7.0	14.0	18.5	17.6	16.0	12.4
Hansen *p*-value	.81	.73	.54	.33	.44	.41	.25	.50
Observations	4,273	4,901	5,517	6,153	5,292	5,292	5,292	5,218
Countries in sample	93	109	131	174	138	138	138	138
	C. HHK Estimates							
Democracy	.222	.234	.144	1.619	1.101	.887	.790	1.713
	(.379)	(.401)	(.445)	(.477)	(.686)	(.679)	(.638)	(.584)
Interaction	.004	−.000	.001	.002	.093	.089	.058	.016
	(.003)	(.003)	(.004)	(.004)	(.046)	(.037)	(.028)	(.013)
Long-run effect of democracy	7.692	7.453	4.480	48.375	31.605	25.022	22.375	49.338
	(13.442)	(13.213)	(14.002)	(21.975)	(21.502)	(20.748)	(19.522)	(23.950)
Effect of democracy after 25 years	4.869	5.084	3.054	34.304	23.787	19.159	17.091	36.069
	(8.286)	(8.850)	(9.435)	(11.965)	(15.084)	(14.981)	(14.107)	(14.116)
Persistence of GDP process	.971	.969	.968	.967	.965	.965	.965	.965
	(.009)	(.008)	(.008)	(.009)	(.008)	(.008)	(.008)	(.009)
Observations	4,180	4,792	5,386	6,110	5,154	5,154	5,154	5,154
Countries in sample	93	109	131	174	138	138	138	138

NOTE.—This table presents estimates of the effect of democracy on log GDP per capita and its interaction with other country characteristics indicated in the column headers. The reported coefficients of democracy and the interaction are multiplied by 100. We report main effects and long-run effects evaluated at the 25th percentile of the interacted variable. Panel A presents within estimates. Panel B presents 2SLS estimates instrumenting democracy (and the interaction term) with four lags of regional democracy waves. It also reports the *F*-statistic for the excluded (excl.) instruments and the *p*-value of Hansen's overidentification test. Panel C presents results using the HHK (Hahn et al. 2001) estimator instrumenting democracy (and the interaction term) with four lags of regional democracy waves. In all specifications, we control for a full set of country and year fixed effects and four lags of GDP per capita. Standard errors robust against heteroskedasticity and serial correlation at the country level are in parentheses.

Figure 23: Original Table 8: Heterogeneous effects